TABLE OF CONTENTS

Page

7. LAND CONTAMINATION 7-1

7.1 Introduction 7-1
7.2 Environmental Legislation, Standards, and Guidelines 7-1
7.3 Description of the Environment 7-1
7.4 Assessment Methodology 7-2
7.5 Identification of Potential Land Contamination Concern 7-2
7.6 Site Investigation Plan 7-37
7.7 Prediction and Evaluation of Environmental Impacts 7-42
7.8 Mitigation of Adverse Environmental Impacts 7-43
7.9 Evaluation of Residual Impacts 7-52
7.10 Environmental Monitoring and Audit 7-53
7.11 Conclusion 7-53

List of Table

Table 7.1	Summary of Findings in Previous EIAs
Table 7.2	Summary of Historical Land Uses within the Project Site
Table 7.3	Summary of Chemical Waste Producer Records
Table 7.4	Summary of Issued Dangerous Goods License
Table 7.5	Summary of Incident Records within Project Site
Table 7.6	Details of Site Appraisal for Potentially Contaminated Sites and Sites with Suspected Industrial Land Uses
Table 7.7	Preliminary Sampling and Testing Plan
Table 7.8	Possible Remediation Methods for Potentially Contaminating Land Uses
Table 7.9	Possible Remediation Methods for Soil and Groundwater Contamination
Table 7.10	Tentative Schedule of Submission

List of Figures

Figure 7.1	Location of Arsenic Areas

Appendices

Appendix 7.1	Extract of previous EIA Reports
Appendix 7.2	Selected Aerial Photographs
Appendix 7.3	Correspondence with Environmental Protection Department
Appendix 7.4	Correspondence with Fire Services Department and Summary of Incident Record
Appendix 7.5	Photographic Records of Site Walkover
Appendix 7.6	Site Walkover Checklist
Appendix 7.7	Risk-Based Remediation Goals (RBRGs) for Soil and Soil Saturation Limit and for Groundwater and Solubility Limit

7. LAND CONTAMINATION

7.1 Introduction

7.1.1.1 This section presents the findings of evaluation and assessment on the potential land contamination issues within the Project Site. The land contamination issues have been assessed in accordance with the requirement in Annex 19 of the Technical Memorandum on Environmental Impact Assessment Process (EIAO-TM) and the requirements in Section 3.5.7 and Appendix F of the EIA Study Brief (ESB-373/2025).

7.2 Environmental Legislation, Standards, and Guidelines

7.2.1.1 The relevant environmental legislation guidelines and standards on land contamination aspect include the following:

- Section 3 (Potential Contaminated Land Issues) of Annex 19 "Guidelines for Assessment of Impact on Sites of Cultural Heritage and Other Impacts" of the Technical Memorandum on Environmental Impact Assessment Process (EIAO-TM);

- Guidance Note for Contaminated Land Assessment and Remediation, EPD, April 2023 (Guidance Note): This Guidance Note sets out the requirements for proper assessment and management of potentially contaminated sites such as oil installations (e.g. oil depots, petrol filling stations), gas works, power plants, shipyards/boatyards, chemical manufacturing/processing plants, steel mills/metal workshops, car repairing/dismantling workshops and scrap yards. In addition, this Guidance Note provides guidelines on how site assessments should be conducted and analysed and suggests practical remedial measures that can be adopted for the remediation of contaminated sites;

- Practice Guide for Investigation and Remediation of Contaminated Land, EPD, April 2023 (Practice Guide): This Practice Guide outlines typical investigation methods and remediation strategies for the range of potential contaminants typically encountered in Hong Kong; and

- Guidance Manual for Use of Risk-Based Remediation Goals (RBRGs) for Contaminated Land Management, EPD, April 2023 (Guidance Manual): This Guidance Manual introduces the risk-based approach in land contamination assessment and presents instructions for comparison of soil and groundwater data to the RBRGs for 54 chemicals of concern (COCs) commonly found in Hong Kong. The RBRGs were derived to suit Hong Kong conditions by following the international practice of adopting a risk-based methodology for contaminated land assessment and remediation, and were designed to protect the health of people who could potentially be exposed to land impacted by chemicals under four broad post restoration land use categories. The RBRGs also serve as the remediation targets if remediation is necessary.

7.3 Description of the Environment

7.3.1.1 The location and boundary of the Project Site is shown in **Figure 1.1**. The Assessment Area for land contamination assessment comprises areas within the Project boundary and any associated works. Existing land use within the Project

includes agricultural land, fish ponds, rural villages, hillside and diverse brownfield land. Brownfield land refers to former agricultural land converted into various uses including open storage, warehousing, workshops, logistics and freight operations. Due to the varied nature of potentially contaminative operations on brownfield land, these types of sites have greater potential for land contamination issues to exist.

7.4 Assessment Methodology

7.4.1.1 Land contamination assessment was carried out according to the EIAO-TM, Practice Guide, Guidance Note and Guidance Manual as well as the requirements given in Section 3.5.7 and Appendix F of the EIA Study Brief (ESB-373/2025).

7.4.1.2 Site appraisals, comprising site walkovers and desktop review, were conducted to identify the potentially contaminating land uses that may pose adverse impact to the Project. Site walkovers were conducted within the Project Site to identify any existing contaminative land uses and contamination sources (or 'hotspots'). For the desktop review, the following information was reviewed:

- Geotechnical Assessment Report under the Project (Ref. R09);
- Hong Kong Geological Survey (HKGS) Map (Series HGM20) – Sheet No. 2 (1:20,000); and
- Selected aerial photographs and topographic maps held by the Lands Department.
- Dangerous Goods and chemical spillage/leakage incidents records. - Fire Services Department (FSD).
- Chemical waste producers register and chemical spillage/leakage incidents - EPD.

7.5 Identification of Potential Land Contamination Concern

7.5.1 Review of Previous EIA Studies

7.5.1.1 The following relevant EIA studies were reviewed. Relevant sections from the EIA reports are extracted and provided in **Appendix 7.1**.

- EIA Report – Hong Kong Section of Guangzhou - Shenzhen - Hong Kong Express Rail Link (Register No.: AEIAR-143/2009)
- EIA Report – The North East New Territories New Development Areas Environmental Impact Assessment (NENT NDA EIA) (EIA Register No. AEIAR-175/2013)
- EIA Report - Northern Link (EIA Register No. AEIAR-259/2024)
- EIA Report – First Phase Development of the New Territories North – San Tin / Lok Ma Chau Development Node – Investigation (EIA Register No. AEIAR-261/2024)

- EIA Report – Development at Ngau Tam Mei Area (EIA Register No. AEIAR-271/2025)

7.5.2 Relevant findings from these EIA reports are summarised in the following sections.

EIA Report – Hong Kong Section of Guangzhou - Shenzhen - Hong Kong Express Rail Link (Register No.: AEIAR-143/2009)

7.5.2.1 A land contamination assessment was conducted under the approved EIA report. The assessment was conducted in accordance with the EIAO-TM as well as EPD's guidelines at the time of the report. The EIA report was approved in September 2009. The assessment included desktop study and site inspections which were conducted in 2008.

7.5.2.2 The only relevant section for this Project is the works area for Ngau Tam Mei Ventilation Building under Express Rail Link. According to the findings of the approved EIA report, there were no potentially contaminated sites encroached into the Project.

EIA Report – The North East New Territories New Development Areas Environmental Impact Assessment (NENT NDA EIA) (EIA Register No. AEIAR-175/2013)

7.5.2.3 A land contamination assessment was conducted under the approved EIA report. The assessment was conducted in accordance with the EIAO-TM as well as EPD's guidelines at the time of the report. The EIA report was approved in October 2013. The assessment included desktop study and site inspections which were conducted in 2009 and 2012.

7.5.2.4 According to the approved EIA report, 1 site of "A" rating - high potential on land contamination, 12 sites of "B" rating – low potential on land contamination, and 1 site of "U" rating - limited for access for visual inspection within the assessment area would encroach into the Project Site at San Tin.

EIA Report - Northern Link (EIA Register No. AEIAR-259/2024)

7.5.2.5 A land contamination assessment was conducted under the approved Northern Link EIA report. The assessment was conducted in accordance with the EIAO-TM as well as EPD's guidelines at the time of the report. The EIA report was approved in February 2024. The assessment included desktop study and site surveys which were conducted in 2022 and 2023.

7.5.2.6 According to the approved EIA report, 16 potentially contaminated sites and 2 sites with no land contamination potential identified within the assessment area would encroach into the Project Site at San Tin and Fung Kat Heung.

7.5.2.7 Concerning Open storage (NTM027 / NMH-ST-066), with further site appraisal and aerial photo review, the site had been a farm since 1990s. Land contamination is not anticipated.

EIA Report – First Phase Development of the New Territories North – San Tin / Lok Ma Chau Development Node – Investigation (EIA Register No. AEIAR-261/2024)

7.5.2.8 A land contamination assessment was conducted under the approved San Tin / Lok Ma Chau Development Node EIA report. The assessment was conducted in accordance with the EIAO-TM as well as EPD's prevailing guidelines. The EIA report was approved in May 2024. Drone reconnaissance and site visits were conducted in 2022 and 2023.

7.5.2.9 Based on the approved EIA report, 2 potentially contaminated sites, 1 site of uncertain nature of operations and 5 sites with no land contamination potential identified within the assessment area would encroach into the Project Site at San Tin.

EIA Report – Development at Ngau Tam Mei Area (EIA Register No. AEIAR-271/2025)

7.5.2.10 A land contamination assessment was conducted under the approved Development at Ngau Tam Mei EIA report. The assessment was conducted in accordance with the EIAO-TM as well as EPD's prevailing guidelines. The EIA report was approved on 18 December 2025. Drone reconnaissance and site visits were conducted in 2022 and 2023.

7.5.2.11 Based on the approved EIA report, 5 potentially contaminated sites and 1 site of uncertain nature of operations were identified within the assessment area would encroach into the Project Site at Ngau Tam Mei.

7.5.2.12 A summary of findings is presented in **Table 7.1**.

Table 7.1 Summary of Findings in Previous EIAs

Site ID in this EIA Report	Usage and Site ID in Previous EIAs	Classified as Potentially contaminated site in the previous EIA?
NMH-ST-001	Vacant grassland (previously as quarry) (KTN-91) [1]	"B" - Low Potential [1]
NMH-ST-002	Trailer parking area (paved, small scale) (KTN-89) [1]	"B" - Low Potential [1]
NMH-ST-003	Grassland with few temporary structures (KTN-88a) [1]	"B" - Low Potential [1]
NMH-ST-004	Brick Manufacturing (partially paved) (KTN-87) [1]	"B" - Low Potential [1]
NMH-ST-006	MTR Spur Line Emergency Access Point No.5. (KTN-87a) [1]	"B" - Low Potential [1]
NMH-ST-007	Vehicle maintenance (PAA002) [2] Warehouse (PAA003) [2]	Yes [7]
	Suspected workshop / storage (partially paved) (KTN-87b) [1]	"B" - Low Potential [1]
NMH-ST-008	Vehicle maintenance (PAA002) [2]	Yes [7]
	Vehicle maintenance (PAA004) [2]	Yes

Site ID in this EIA Report	Usage and Site ID in Previous EIAs	Classified as Potentially contaminated site in the previous EIA?
	Suspected "Pig-oil" manufacturer (Partially paved) (KTN-87c) [1]	"B" - Low Potential [1]
NMH-ST-009	Open carpark and office (PAA006) [2]	No [6]
NMH-ST-010	Open carpark (previous workshop) (PAA005) [2]	Yes
NMH-ST-011	Scrapyard. (E055) [3]	Yes
NMH-ST-012	Vacant land. (E056) [3]	No
NMH-ST-013	Electricity substation (E057) [3]	TBC [5]
NMH-ST-014	Vehicle workshop (E069) [3]	Yes
NMH-ST-015	Vehicle repair and maintenance workshop (small scale, paved) (KTN-86) [1]	"B" - Low Potential [1]
NMH-ST-016	Storage warehouse (KTN-84a) [1]	"U" - limited for access for visual inspection during the survey [1]
NMH-ST-017	Shun Hai, Trailer Park, Repair and Maintenance Activities (paved) (KTN-84) [1]	Yes
NMH-ST-018	Vacant (partially paved) (KTN-82 and KTN-83) [1]	"B" - Low Potential [1]
NMH-ST-022	Car storage (S063) [3]	No
NMH-ST-023	Pig Farm (S066) [3]	No
NMH-ST-024	Pig Farm (S065) [3]	No
NMH-ST-025	Industrial land use, suspected construction materials / equipment storage (S63) [4]	Yes
NMH-ST-026	Construction materials / equipment storage. (S52) [4]	Yes
NMH-ST-027	Open storage of vehicles and construction materials (S43) [4]	Yes
NMH-ST-039	Warehouse (AUT012) [2]	Yes [7]
NMH-ST-045	Open storage and residential (AUT002) [2]	Yes
NMH-ST-046	Workshop (AUT008) [2]	Yes [7]
NMH-ST-047	Open Storage (AUT003) [2]	No [6]
NMH-ST-048	Warehouse (AUT004) [2]	Yes [7]
NMH-ST-049	Warehouse (AUT005) [2]	Yes
NMH-ST-054	Warehouse (PAA001) [2]	Yes [7]
NMH-ST-055	Warehouse (LHA008) [2]	Yes [7]
NMH-ST-056	Warehouse (LHA001) [2]	Yes
NMH-ST-059	Tung Hing Metal Workshop/ Storage (KTN-88b) [1]	"B" - Low Potential [1]
NMH-ST-060	Abandoned pig farm (KTN-85) [1]	"B" - Low Potential [1]

Site ID in this EIA Report	Usage and Site ID in Previous EIAs	Classified as Potentially contaminated site in the previous EIA?
NMH-ST-062	Abandoned pig farm (S064) [3]	No
NMH-ST-063	Site Depot (S47) [4]	Yes
NMH-ST-064	Vehicle maintenance workshop and vehicles storage (S45) [4]	Yes
NMH-ST-065	Industrial land use, temporary structures and suspected construction materials (S42) [4]	TBC [5]
NMH-ST-066	Open storage (NTM027) [2,6,8]	Yes [7]
NMH-ST-067	Open storage (NTM026) [2,]	Yes [7]
NMH-ST-068	Recycling facilities (NTM011) [2]	Yes

Note:
1 EIA Report - The North East New Territories New Development Areas Environmental Impact Assessment (NENT NDA EIA) (EIA Register No. AEIAR-175/2013)
2 EIA Report - Northern Link (EIA Register No. AEIAR-259/2024)
3 EIA Report – First Phase Development of the New Territories North – San Tin / Lok Ma Chau Development Node – Investigation (EIA Register No. AEIAR-261/2024)
4 EIA Report - Development at Ngau Tam Mei Area (EIA Register No. AEIAR-271/2025)
5 TBC: To be confirmed
6 No potential contamination source
7 The site was inaccessible and no drillhole was proposed. Site re-appraisal was recommended.
8 Further site appraisal and aerial photo review confirmed the site (NTM027) had been a farm since 1990s. Land contamination is not anticipated.

7.5.3 Review of Historical Land Uses

7.5.3.1 A review of historical aerial photographs covering the Project Site has been undertaken. The aim of the review is to identify any historical land uses within the Project Site that may have potential contamination implications to the Project. A summary of the historical land uses is listed in and the selected aerial photographs are provided in **Appendix 7.2**.

Table 7.2 Summary of Historical Land Uses within the Project Site

Year	Drawing No. in Appendix 7.2	Description of Land Uses and Site Operation/Activities within Project Site
1963	Aerial Photo 1.01 – 1.18	San Tin * • The majority of the Project area comprised natural terrain and agricultural land. Village development is observed. Ngau Tam Mei • The majority of the Project area comprised natural terrain and agricultural land. Village development is observed. Mo Fan Heung / Fung Kat Heung • The majority of the Project area comprised natural terrain and agricultural land. Village development is observed. • Ponds are observed at the southern end.

Year	Drawing No. in Appendix 7.2	Description of Land Uses and Site Operation/Activities within Project Site
1973	Aerial Photo 2.01 – 2.18	San Tin • Some natural terrain and agricultural land was changed to developed land. Ngau Tam Mei • No significant change in land use in comparison to 1963. Mo Fan Heung / Fung Kat Heung • No significant change in land use in comparison to 1963. • The Soy Sauce Factory (NMH-ST-050) was first observed and some ponds was observed near the factory.
1982	Aerial Photo 3.01 – 3.18	San Tin • More developed area (warehouse/storage) was observed. • A large area in Kwu Tung was vacated. • Two farms in (NMH-ST-023 & NMH-ST-024) are being established. Ngau Tam Mei • Some agricultural land was changed to ponds. Mo Fan Heung / Fung Kat Heung • No significant change in land use in comparison to 1973.
1993	Aerial Photo 4.01 – 4.18	San Tin • More developed area (warehouse/storage) was observed. • San Tin Highway was widened. Ngau Tam Mei • No significant change in land use in comparison to 1982. Mo Fan Heung / Fung Kat Heung • Some pond was filled to land for warehouse/storage. • San Tin Highway was widened.
2002	Aerial Photo 5.01 – 5.18	San Tin • More developed area (warehouse/storage/trailer park) was observed. Ngau Tam Mei • More village development was observed. • Ngau Tam Mei Water Treatment Works was in place. Mo Fan Heung / Fung Kat Heung • More pond was filled to land for warehouse/storage. • Tsing Long Highway was in place. • Kam Tin River was widened.
2010	Aerial Photo 6.01 – 6.18	San Tin • No significant change in land use in comparison to 2002. Ngau Tam Mei • Some developed area (warehouse/storage) was observed. • Construction of ventilation building of Express Rail Link at Ching Yau Road Mo Fan Heung / Fung Kat Heung • More warehouse/storage was observed.

Year	Drawing No. in Appendix 7.2	Description of Land Uses and Site Operation/Activities within Project Site
2014	Aerial Photo 7.01 – 7.18	San Tin • No significant change in land use in comparison to 2010. Ngau Tam Mei • No significant change in land use in comparison to 2010. Mo Fan Heung / Fung Kat Heung • No significant change in land use in comparison to 2010. • More pond was filled to land for warehouse/storage.
2017	Aerial Photo 8.01 – 8.18	San Tin • No significant change in land use in comparison to 2014. Ngau Tam Mei • No significant change in land use in comparison to 2014. • Some large car storage (NMH-ST-027, NMH-ST-063 and NMH-ST-064) was established. Mo Fan Heung / Fung Kat Heung • No significant change in land use in comparison to 2014. • Some small development near Long Ha (NMH-ST-029, NMH-ST-030, NMH-ST-055 and NMH-ST-056)
2020	Aerial Photo 9.01 – 9.18	San Tin • No significant change in land use in comparison to 2017. • Development of Kwu Tung North New Development Areas commenced. The first development was Kwu Tung North Multi-welfare Services Complex. Ngau Tam Mei • No significant change in land use in comparison to 2017. Mo Fan Heung / Fung Kat Heung • No significant change in land use in comparison to 2017.
2023	Aerial Photo 10.01 – 10.18	San Tin • Major development was observed in Kwu Tung as Kwu Tung North New Development Areas. • Interchange at Pak Shek Au was under modification works. Ngau Tam Mei • No significant change in land use in comparison to 2020. Mo Fan Heung / Fung Kat Heung • No significant change in land use in comparison to 2020. • Previous open storage at NMH-ST-053 was turned into school (Anchors Academy Affiliated International Kindergarten cum Nursery)

Note:
* aerial photo of year 1963 is not available for some area at the northern end at San Tin.

7.5.3.2 Based on the review of aerial photographs, most of the area remained rural with village development, agricultural land or pond until 1990s. Open storage and warehouse began to appear and expand in San Tin and Mo Fan Heung / Fung Kat Heung area since 1990s, while the Ngau Tam Mei area largely remained as ponds with occasional open storage.

7.5.4 Review of Site Geology

7.5.4.1 The superficial deposits in the area primarily consist of fill, colluvium, and alluvium. In some boreholes, only soil from in-situ decomposition was encountered. Fill and

alluvium are found beneath gentle terrain, including Ngau Tam Mei and the Fanling Highway. The thickness of fill varies up to 3 m, while the thickness of alluvium generally ranges from 1 m to 5.5 m. Colluvium is occasionally present at the foothills along the alignment, with thicknesses typically ranging from 3 m to 4 m or less, though it can reach an thickness of approximately 24 m in some locations. The alignment crosses various faults, which may result in abnormal thickness of certain superficial deposits.

7.5.4.2 The solid geology along the alignment varies. Based on the 2nd edition 1:100,000 geological map, the majority of the San Tin section is underlain by the Tai Mo Shan Formation (Jtm), which typically consists of pale-grey to dark-grey, lapilli- and lithic-bearing ash or coarse ash crystal tuff. The lapilli- and lithic-bearing coarse crystal tuff of the Shing Mun Formation (Jts) is locally encountered in the Nam Sang Wai and Sha Po Tsuen areas.

7.5.4.3 In the Chau Tau area, the alignment is underlain by slightly metamorphosed grey to red fine sandstone, laminated siltstone, and black carbonaceous mudstone of the Tai O Formation (Jo). A NE-SW trending San Tin thrust fault intersecting the alignment has caused the Tai O Formation (Jo) to be thrust over the Tai Mo Shan Formation (Jtm). Therefore, it may be inferred that a tuffaceous igneous basement underlies the metamorphic and meta-sedimentary rocks.

7.5.5 Information from Relevant Government Departments

7.5.5.1 EPD and FSD were contacted for (i) records on any release of chemicals and chemical waste; (ii) records of DG; (iii) records of CWP(s); and (iv) records of reported incidents within the Project Site. Replies from EPD and FSD are provided in **Appendices 7.3** and **7.4** respectively. The information is summarised below.

7.5.6 Environmental Protection Department

7.5.6.1 Based on the reply given by EPD on 8 April 2025 and 26 November 2025, there is no record of spillage / leakage of chemical wastes or chemicals found within the Project Site. In addition, visits to EPD's Territorial Control Office were undertaken on 12 August and 16 September 2025 to review the available CWP records. There is a total of 11 valid and 3 invalid registered CWP records relevant to the Project Site. Details of the CWP records are summarised in **Table 7.3**.

Table 7.3 Summary of Chemical Waste Producer Records

No.	Site ID[1]	Company Name	Address	Status	Business Nature	Within Project Site?[1]
1	NMH-ST-009	GOODPLACE ENVIRONMENTAL PROTECTION LIMITED	DD96 Lot 621, Yuen Long, NT	Invalid	環保處理回收廢電子元件，廢五香及廢塑料	Yes
2	NMH-ST-010	THE JARDINE ENGINEERING CORPORATION LMITED	DD96 Lot 2312, Yuen Long, NT	Valid	Eng'g Contractors & Eng. Mer'ts	Yes
3	NMH-ST-011	Wai Ko Recycle Trading Limited	Godown DD96 Lot 497RP, Chau Tau Tsuen, San Tin, Yuen Long, NT	Valid	回收	Yes
4	NMH-ST-043	CLOROX HONG KONG LIMITED	LOT 214A IN DD107 MO FAN HEUNG YUEN LONG NT	Valid	WAREHOUSE	Yes
5	NMH-ST-044	WAN Kwok-wang	No. 107, Mo Fan Heung, Yuen Long, N.T. (DD107 Lot 218 S.A.SS.1 & 219 S.A.SS.1)	Valid	Repairing of motor vehicle	Yes
6	NMH-ST-046	Zhentong (HK) Steel Structure Equipment Limited	DD107 Lot 382, 418RP, 419S.A SS. 1RP, 420 S.C. RP, 420 S.B. RP, Fung Kat Heung, Yuen Long, N.T.	Valid	混凝土攪拌機器保養維修	Yes
7	NMH-ST-050	Koon Chun Hing Kee Soy & Sauce Fty Ltd	Lot 1910 D.D. 107, Au Tau Yuen Long NT	Valid	Manufacturing of Sauce Products	Yes
8	NMH-ST-050	CR Construction Company Limited	Lot 1910 RP in DD107, Au Tau, Yuen Long, N.T.	Valid	Construction	Yes
9	NMH-ST-051	OCEAN WORLD ENTERPRISE LIMITED	DD107 LOT 1743 S.C RP, CASTLE PEAK ROAD - TAM MI, NAM SANG WAI, YUEN LONG, N.T.	Valid	回收二手五金，金屬，電腦	Yes
10	NMH-ST-052	China Geo Engineering Corporation	Nam Sang Wai Road, adjacent to Tin Kwong Hui Vegetable Wholesale Market, Yuen Long, NT	Valid	Slope Upgrading Work	Yes

No.	Site ID(1)	Company Name	Address	Status	Business Nature	Within Project Site?(1)
11	NMH-ST-052	China Geo-Engineering Corporation	Government Land Adjoining Nam Sang Wai Road, Yuen Long, GLA-TYL 2957	Valid	Landslip Prevention and Mitigation Works	Yes
12	NMH-ST-053	Build King Construction Limited.	DD103 Lot 1066, Au Tau, Yuen Long, N.T.	Invalid	Construction	Yes (partial encroachment)
13	NMH-ST-053	CR Construction Company Limited	Main Contract Works For Proposed Primary School and Kindergarten Development at Lot 1928 in DD107, Au Tau, Yuen Long, NT.	Valid	Construction	Yes
14	NMH-ST-053	Simon & Sons Engineering Limited	DD107 LOT 1928, Ko Po Path, Au Tau, Yuen Long. NT.	Invalid	Construction & Engineering	Yes

Note:
(1) Locations of the sites (the relevant CWPs) are provided in **Appendix 7.5**.

7.5.6.2 As shown in **Table 7.3**, 14 CWP records (viz. CWP Nos. 1-14) were associated with 10 identified sites (viz. Sites NMH-ST-009, NMH-ST-010, NMH-ST-011, NMH-ST-043, NMH-ST-044, NMH-ST-046, NMH-ST-050, NMH-ST-051, NMH-ST-052 and NMH-ST-053) within the Project Site. CWP No. 8 (NMH-ST-050), CWP No. 10 & 11 (NMH-ST-052) and CWP No. 12, 13 & 14 (NMH-ST-053) were CWP for general construction and do not have potential land contamination.

7.5.6.3 CWP No. 4 (NMH-ST-043) was a CWP for warehouse. The warehouse was operated by CLOROX HONG KONG LIMITED, a home care product company whose product range includes detergents and bleach. It should be noted that certain home care products, such as bleach containing sodium hypochlorite and alkaline cleaners containing sodium hydroxide, are classified as chemical waste when disposed of, for example due to wastage or expiry. This CWP did not indicate potential land contamination. CWP No. 7 (NMH-ST-050) was a CWP for a sauce factory and potential land contamination is not anticipated.

7.5.6.4 CWP No. 1 (NMH-ST-009), CWP No. 2 (NMH-ST-010), CWP No. 3 (NMH-ST-011), CWP No. 5 (NMH-ST-044), CWP No. 6 (NMH-ST-046) and CWP No. 9 (NMH-ST-051) were CWP for maintenance or recycling, which may lead to potential land contamination.

7.5.7 Fire Services Department

7.5.7.1 Based on FSD's replies on 24 April 2025, a total of 35 DG licenses have been issued in the vicinity of the Project and details are summarised in **Table 7.4** below.

Table 7.4 Summary of Issued Dangerous Goods License

No.	Type of DG	Quantity	Location of Storage	Within Project Site?
1	Acetylene, Dissolved	320 litres	Construction Site of Public Housing Development at Kwu Tung North Area 19, Phase 1A and Phase 1B, Kwu Tung, N.T.	No
2	Oxygen, Compressed	752 litres		
3	Gasoline	400 litres		
4	Diesel	22,692 litres	Sinopec Petrol Filling Station, Pok Wai, Lots 3808 & 3809B in D.D. 104, Yuen Long, N.T.	No
5	Petrol	22,692 litres		
6	Petrol	22,692 litres		
7	Diesel	15,000 litres	Sinopec Petrol Filling Station, Fairview Park Boulevard, DD 104, Lot 3250 s. A ss. 1 R.P., Yuen Long, N.T.	No
8	Petrol	8,500 litres		
9	Petrol	11,800 litres		
10	Diesel	22,500 litres	Sinopec LPG cum Petrol Filling Station, Lot No. 1041 in D.D.103, Kam Tin Road, Au Tau, Yuen Long, N.T.	No
11	Diesel	22,500 litres		
12	Petrol	22,500 litres		
13	Petrol	22,500 litres		
14	Diesel	(1) 2,000 litres (2) 2,400 litres (3) 2,000 litres (4) 1,800 litres	Park YOHO, Phase 2, 18 Castle Peak Road, Tam Mi, N.T.	No

No.	Type of DG	Quantity	Location of Storage	Within Project Site?
15	Diesel	(1) 18,184 litres (2) 450 litres (3) 450 litres	Lot 1910, DD 107 Au Tau, Yuen Long, N.T.	Yes (NMH-ST-050)
16	Diesel	(1) 450 litres (2) 5,896 litres		
17	Diesel	30,000 litres	ESSO Fairview Park Filling Station, Lot 4665, D.D.104, Tai Shang Wai, Yuen Long, N.T.	No
18	Petrol	15,000 litres		
19	Petrol	15,000 litres		
20	Petrol	20,000 litres		
21	Petrol	18,000 litres	Caltex Fairview Park Filling Station, Lot 2220RP, 2221RP, 2292B, 2292RP, 2293D, 2293E, 2293RP and part of 2217, 2213, 2212 in DD 104, Yuen Long, N.T.	No
22	Petrol	18,000 litres		
23	Diesel	22,500 litres		
24	Diesel	22,500 litres		
25	Petrol	20,000 litres	Caltex Petrol Filling Station, Lot 3532, C.R.P. in DD 104, Long Ha, Yuen Long, N.T.	No
26	Petrol	20,000 litres		
27	Diesel	22,500 litres		
28	Diesel	22,500 litres		
29	Oxygen, Refrigerated Liquid	13,000 litres	Ngau Tam Mei Water Treatment Works, Chuk Yuen Road, Yuen Long	No
30	Oxygen, Refrigerated Liquid	13,000 litres		
31	Potassium Permanganate	27,000 kg		
32	Diesel	1,300 litres		
33	Diesel	1,300 litres		
34	Sodium Hydroxide Solution	9,500 litres		
35	Sodium Hydroxide Solution	9,500 litres		

7.5.7.2 The only DG licenses located within the Project Site were DG licence No. 15 and 16. The two licenses was located within the Sauce Factory (identified as site NMH-ST-050) at Lot 1910, DD 107. The DG were diesel storage on site.

7.5.7.3 Based on FSD's replies on 24 April 2025, 5 January 2026 and 27 February 2026, 168 individual incidents were recorded by FSD in **Appendix 7.4**. 58 of the recorded incidents are considered within the project site. The details of the incident records are summarised in **Table 7.5**.

Table 7.5 Summary of Incident Records within Project Site

No.	Address / Lamp Post No.	Date	Type of Incident	Within Project Site?
FSD26	Hillside of \UT JK998890 of Ngau Tam Shan	13/4/2024	Mountain Rescue	Yes
FSD40	Hillside \UT KK009903 of Ki Lun Shan	17/1/2024	Mountain Rescue	Yes
FSD49	Hillside \UT KK008906 of Ki Lun Shan	23/10/2023	Mountain Rescue	Yes
FSD61	Hillside near Lamppost No. V3911 of Ngau Tam Mei	13/3/2023	Vegetation Fire	Yes
FSD64	Near Lamppost No. VD2271 Ngau Tam Mei	10/3/2023	Vegetation Fire	Yes
FSD75	Near hillside \UT KK010907 of Kwu Tung Reservoir	11/10/2022	Mountain Rescue	Yes
FSD83	Hillside \UT KK009910 of Ki Lun Shan	23/6/2022	Mountain Rescue	Yes
FSD87	Near Lamppost No. FA4043 of Ki Lun Tsuen	26/4/2022	Vegetation Fire	Yes
FSD90	Junction of San Tam Road and Fung Kat Heung Road	19/10/2025	Rubbish Fire	Yes
FSD91	Hillside, UTM Grid Reference JK961855, Near Ho Hok Shan	15/10/2025	Mountain Rescue	Yes
FSD92	Near Lamppost FC3658, Near Pok Wai South Road	30/8/2025	Vehicle Fire	Yes
FSD93	Lamppost EA2204, Near Fanling Highway, Yuen Long Bound, Near Kwu Tung	25/8/2025	Special Service: Fuel leakage from a vehicle constituting fire hazard	Yes
FSD94	Near Lamppost EA1995, Fanling Highway, Yuen Long Bound, Near Kwu Tung	25/8/2025	Traffic Accident	Yes
FSD95	Roadside, Near Lamppost EA2024, Near Kwu Tung Road	12/8/2025	Vehicle Fire	Yes
FSD96	Hillside, Lamppost V3911, Ngau Tam Mei	28/7/2025	Vegetation Fire	Yes
FSD97	Near Lamppost FB9523, Near Tsing Long Highway, Yuen Long Bound. Near Fung Kat Heung	21/6/2025	Traffic Accident	Yes
FSD99	Near Lamppost FB9523, Near Tsing Long Highway, Yuen Long Bound, Near Fung Kat Heung	13/6/2025	Traffic Accident	Yes
FSD100	Un-numbered House, Near Lamppost VD6532, Yau Tam Mei	7/6/2025	Lock-in Case	Yes
FSD102	Hillside, UTM Grid Reference KK009903, Ki Lun Shan	4/4/2025	Mountain Rescue	Yes
FSD103	Hillside Lamppost FC2949, Near Ching Yau Road	4/4/2025	Vegetation Fire	Yes
FSD104	Hillside Lamppost H3346, Kei Lun Tsuen	3/4/2025	Vegetation Fire	Yes
FSD106	Near Lamppost FB9522, Near San Tin Highway, Sheung Shui Bound, Near Au Tau	19/3/2025	Vegetation Fire	Yes
FSD108	Hillside, Near Lamppost FA9260, Near San Tam Road	12/1/2025	Vegetation Fire	Yes

No.	Address / Lamp Post No.	Date	Type of Incident	Within Project Site?
FSD111	Kam Tin River, Near Lamppost BD0380, Nam Sang Wai Road	8/11/2024	Drown Case	Yes
FSD112	Lamppost FB9526, Near Tsing Long Highway, Kowloon Bound, Near Fung Kat Heung	30/10/2024	Vegetation Fire	Yes
FSD115	Roadside, Near Lamppost FC3658, Near Pok Wai South Road	8/5/2024	Vehicle Fire	Yes
FSD116	Near Lamppost GD2979, Near San Tam Road	21/3/2024	Rubbish Fire	Yes
FSD117	Near Lamppost FB9532, Near Au Tau, Near Tsing Long Highway, New Territories Bound	21/3/2024	Vegetation Fire	Yes
FSD118	Near Lamppost FB9569, Near Pok Oi Hospital, Near Yuen Long Highway, Tuen Mun Bound	21/3/2024	Vegetation Fire	Yes
FSD120	Near Lamppost FB9541, Near Castle Peak Road – Tam Mei	4/3/2024	Vegetation Fire	Yes
FSD121	Near Lamppost FA9256, Near San Tam Road	22/2/2024	Vegetation Fire	Yes
FSD122	Near Lamppost FB9540, Near Castle Peak Road – Tam Mei	21/1/2024	Vegetation Fire	Yes
FSD123	Unnamed Road, Near UTM Grid Reference JK960855	9/1/2024	Vegetation Fire	Yes
FSD125	Openground, Near Junction of Nam Sang Wai Road and Castle Peak Road – Tam Mei	25/12/2023	No. 1 Alarm Fire	Yes
FSD127	Near Lamppost EA2009, Castle Peak Road – Chau Tau	4/12/2023	Vegetation Fire	Yes
FSD128	Unnumbered House, Near Lamppost VD1845, Ngau Tam Mei	28/11/2023	Late Call Case	Yes
FSD129	Near Lamppost FC3656, Near Pok Wai South Road	14/11/2023	Rubbish Fire	Yes
FSD130	G/F, MTR Lok Ma Chau Spur Line EAP 5, Hung To Road	10/11/2023	No. 2 Alarm Fire	Yes (NMH-ST-006)
FSD131	Near Lamppost FA9255, Near San Tam Road	22/9/2023	Vegetation Fire	Yes
FSD136	Hillside,UTM Grid Reference JK974876, Kai Kung Shan	2/8/2023	Mountain Rescue	Yes
FSD139	Roadside, Near Lamppost FC2749, Near Nam Sang Wai Road	25/4/2023	No. 1 Alarm Fire	Yes
FSD141	Near Lamppost FA9236, Near San Tin Highway – Yuen Long Bound, Near Fung Kat Heung	18/4/2023	Traffic Accident	Yes
FSD142	Near Lamppost AJ0068, Near Nam Sang Wai Road	10/3/2023	Vegetation Fire	Yes
FSD143	Near Lamppost FB9530, Near Tsing Long Highway, Kowloon Bound Near Fung Kat Heung	22/2/2023	Vegetation Fire	Yes
FSD145	Near Lamppost V3146, Near Ngau Tam Mei	17/2/2023	Vegetation Fire	Yes
FSD146	Near Lamppost FB9628, Near Nam Sang Wai Road	17/2/2023	Vegetation Fire	Yes

No.	Address / Lamp Post No.	Date	Type of Incident	Within Project Site?
FSD147	Near Lamppost FB9628, Near Nam Sang Wai Road	12/2/2023	Vegetation Fire	Yes
FSD148	Roadside, Near Slope 6NE-A/F54, Near Yuen Long Highway	26/1/2023	Vegetation Fire	Yes
FSD150	Near Lamppost FB9541, Near Castle Peak Road – Tam Mei	6/1/2023	Vegetation Fire	Yes
FSD151	Near Lamppost FB9527, Near Tsing Long Highway, Kowloon Bound, Near Fung Kat Heung	2/1/2023	Vegetation Fire	Yes
FSD154	Near Lamppost FA9252, Near San Tin Highway, Sheung Shui Bound, Near Au Tau	23/12/2022	Traffic Accident	Yes
FSD156	Openground, Near Lamppost FA8546, Chau Tau	28/1/2026	Rubbish Fire	Yes
FSD157	Near MTR West Rail Line Emergency Access Point 16, Ko Po Path	24/1/2026	Rubbish Fire	Yes (NMH-ST-053)
FSD158	Roadside, Near Lamppost FB9530, Near Tsing Long Highway, New Territories Bound, Near Au Tau	7/1/2026	Special Service: Object Removal	Yes
FSD159	Near Lamppost Y10009, Tsing Long Highway, Kowloon Bound, Near Fung Kat Heung	19/12/2025	Late Call Fire	Yes
FSD162	Hillside, UTM Grid Reference JK969872, Pok Wai	14/1/2025	Vegetation Fire	Yes
FSD167	Roadside, Near Lamppost FA4045, Ki Lun Tsuen	4/11/2023	Rubbish Fire	Yes
FSD168	DD107 Lot 422A, Fung Kat Heung	15/7/2023	No. 1 Alarm Fire	Yes (NMH-ST-049)

7.5.7.4 As shown in **Appendix 7.4**, of the 168 incident records, 107 incidents occurred outside the Project Site and 3 incident location cannot be identified. Thus, these 110 incidents are not expected to pose any potential land contamination issue to the Project.

7.5.7.5 For Incident Record No FSD93, spillage of diesel fuel from a public bus after traffic accident was recorded. The incident report is provided in **Appendix 7.4**. The leaked diesel fuel was covered with sawdust for clean-up. As this was an isolated incident, potential land contamination corresponding to this incident is not anticipated.

7.5.7.6 For Incident Record No FSD130, a No. 2 Alarm Fire was recorded at MTR Lok Ma Chau Spur Line Emergency Access Point 5, which corresponds to the surveyed site NMH-ST-006. The incident report is provided in **Appendix 7.4**. Excessive heat generated from a street hydrant pump ignited its insulating materials and in turn set fire to the same. The fire was put out with carbon dioxide fire extinguisher. With the current operation of emergency access point, it is expected that the fire incident was cleaned up and rectified. As this was an isolated incident, potential land contamination corresponding to this incident is not anticipated.

7.5.7.7 For the remaining 56 incident records, 10 of them are mountain rescue, lock-in case, drown case which do not result in potential land contamination. 46 of them are

vegetation fire, rubbish fire, vehicle fire, traffic accident, No. 1 alarm fire, No. 2 alarm fire, late call fire and late call case. FSD advised that the 46 incidents did not involve any dangerous goods or chemical leakage. Given the nature of the incidents are relative minor with no dangerous goods or chemical leakage, potential land contamination issues associated with these incidents to the Project are not expected. The location of the incidents within the Project Site is provided in **Appendix 7.5**.

7.5.8 Geochemical Atlas of Hong Kong

7.5.8.1 With reference to the *Preliminary Feasibility Study on Developing the New Territories North*, the *Geochemical Atlas of Hong Kong* published by the Geotechnical Engineering Office of Civil Engineering and Development Department in 1999 and the Hong Kong Environmental Database (HKED) by EPD, natural occurrences of elevated level of arsenic exist over broad areas of the northern New Territories. The arsenic area (>571 mg/kg) as extracted from HKED is provided in **Figure 7.1**. A separate arsenic assessment and remediation plan should be prepared and submitted to EPD for endorsement prior to the commencement of construction works.

7.5.9 Site Walkovers

7.5.9.1 Site walkovers with drone survey were conducted on 22 September 2025, 3 October, 8 October, 10 October, 28 November and 11 December 2025 to investigate any land contamination issues associated with the current land uses within the Project Site. Photographic records, along with the site layout plan for the Project Site, are shown in **Appendix 7.5** and the site walkover checklist is provided in **Appendix 7.6**. No current / previous site owner was available for interview.

7.5.9.2 Based on the findings of drone survey and the site walkovers, various suspected industrial activities, such as construction materials / equipment storage, open storage, scrapyard, warehouses and vehicle parking / maintenance workshops were identified within the Project Site. However, detailed site inspection at each individual premise of these sites with suspected industrial activities was restricted these sites are inaccessible during site walkovers (e.g. gate closed / no personnel were available). Hence, only peripheral inspection (i.e. from the entrance and / or boundary of the sites) was conducted for these sites in order to obtain a general view of the Project Site.

7.5.9.3 Findings of the drone survey and site walkovers for the concerned sites are summarised in **Table 7.6** and their locations are shown in **Appendix 7.5**. The corresponding photographic records are shown in **Appendix 7.5**.

7.5.9.4 Based on the drone survey findings and observations during the site walkovers, there were also areas with non-contaminating land uses, including agricultural land, vacant / vegetated land, agricultural and aquaculture farms, fishponds, nullah, village houses / temporary residential structures, village offices, former school, playground/basketball courts, refuse collection points, cycle tracks, roads (i.e. San Tin Highway, San Tam Road, Ngau Tam Mei Road, Castle Peak Road (Chau Tau) and Kwu Tung Road), open car parks, graves and temples within the Project Site. No chemical / oil storage, oil stains or stressed vegetation were observed, and no current potentially contaminating land uses / activities were identified within these areas during the site walkovers.

7.5.10 Summary of Site Appraisals

7.5.10.1 Based on the findings of the site appraisals, a total of 68 surveyed site was recorded. A total of 30 potentially contaminated sites and 15 sites suspected to be used for industrial purposes were identified within the Project Site. The remaining 23 sites are not considered as potentially contaminated sites. Locations of these sites are shown in **Appendix 7.5** and details of these sites are summarised in **Table 7.6**. Intrusive site investigation (SI) works is considered necessary to confirm any land contamination within the potentially contaminated sites. The corresponding sampling and testing strategies are presented in **Section 7.6**.

7.5.10.2 For sites suspected to be used for industrial purposes, further site appraisal should be carried out for the whole Project Site in next stage when site access is available to confirm the existing land uses / activities and to identify the presence of any potential contamination sources (refer to **Sections 7.8.1 to 7.8.1.4** for details).

7.5.11 Future Land Uses

7.5.11.1 The future land use of the whole Project area is defined as "Road". Section 3.1 of Guidance Manual states that the proposed Risk Based Remediation Goals (RBRGs) reflecting the planned future land use "Roads including pedestrian walkway" is "Lower of Industrial or Public Park". Relevant RBRGs are shown in **Appendix 7.7**.

7.5.11.2 In the event that future land use is revised subsequent to approval of this EIA report, RBRGs to be adopted should be updated with reference to the latest proposed land use or proposed facilities.

Table 7.6 Details of Site Appraisal for Potentially Contaminated Sites and Sites with Suspected Industrial Land Uses

Site ID (Affected Lot No. / Address)[1]	Suspected Current Land Use	Approx. Site Area (m^2)	Site Observation and Information from EPD/FSD	Corresponding Photographic Record Reference in Appendix 7.5	Other Past Potentially Contaminating Land Uses / Activities	Corresponding Historical Aerial Photograph Reference in Appendix 7.2	Potentially Contaminated Site?	Potential COCs[3]
NMH-ST-001	Vegetated land	1,418	• The site is located to the north of Pak Shek Au Village. The site was heavily vegetated and inaccessible for detailed site walkover at the time of reporting. • Based on peripheral observations and the drone survey, the site is a vegetated land. • Based on information from the EPD / FSD, there were no DG / chemical wastes / fire incident / spillage records for the site.	60750817/R39.2/801	No past potentially contaminating activities were identified from historical aerial photo. Previous EIA Report (AEIAR-175/2013) identified the site (KTN-91) as grassland (previously as quarry) with "B - Low Potential" on land contamination and recommended for revisit.	Aerial Photo 1.01 – 10.01	No	Not Applicable
NMH-ST-002	Vacant land (unpaved)	714 (Area within Project Site: 643)	• The site is located to the north of Pak Shek Au Village and was accessible for detailed site walkover at the time of reporting. • Based on site observations and the drone survey, the site is a vacant land. No potential land contamination activity/hotspot was identified. • Based on information from the EPD / FSD, there were no DG / chemical wastes / fire incident / spillage records for the site.	60750817/R39.2/802	No past potentially contaminating activities were identified from historical aerial photo. Previous EIA Report (AEIAR-175/2013) identified the site (KTN-89) as trailer park area with "B - Low Potential" on land contamination and recommended for revisit.	Aerial Photo 1.03 – 10.03	No	Not Applicable
NMH-ST-003	Industrial land use[2] - Temporary structures (unpaved)	769 (Area within Project Site: 148)	• The site is located to the north of Pak Shek Au Village and was inaccessible for detailed site walkover at the time of reporting. • Based on peripheral observations and the drone survey, the site was fenced with barrier and some temporary structure was observed. • Based on information from the EPD / FSD, there were no DG / chemical wastes / fire incident / spillage records for the site.	60750817/R39.2/802	No past potentially contaminating activities were identified from historical aerial photo. Previous EIA Report (AEIAR-175/2013) identified the site (KTN-88a) as grassland with few temporary structures with "B - Low Potential" on land contamination and recommended for revisit.	Aerial Photo 1.01 – 10.01	TBC[2]	TBC
NMH-ST-004	Construction Site	883 (Area within Project Site: 434)	• The site is located to the north of Pak Shek Au Village. The site was part of the construction site of Kwu Tung North development and inaccessible for detailed site walkover at the time of reporting. • Based on peripheral observations and the drone survey, the site was part of construction site forming new road. • Based on information from the EPD / FSD, there were no DG / chemical wastes / fire incident / spillage records for the site.	60750817/R39.2/803	No past potentially contaminating activities were identified from historical aerial photo. Previous EIA Report (AEIAR-175/2013) identified the site (KTN-87) as brick manufacturing with "B - Low Potential" on land contamination and recommended for revisit.	Aerial Photo 1.03 – 10.03	No	Not Applicable

Site ID (Affected Lot No. / Address)[1]	Suspected Current Land Use	Approx. Site Area (m^2)	Site Observation and Information from EPD/FSD	Corresponding Photographic Record Reference in Appendix 7.5	Other Past Potentially Contaminating Land Uses / Activities	Corresponding Historical Aerial Photograph Reference in Appendix 7.2	Potentially Contaminated Site?	Potential COCs[3]
NMH-ST-005	Vehicle maintenance workshop (paved)	1,491 (Area within Project Site: 1,489)	• The site is located to the north of Pak Shek Au Village and was inaccessible for detailed site walkover at the time of reporting. • Based on peripheral observations and the drone survey, warehouses and vehicle workshop were observed. Stockpile of car tyres were observed on site. • Based on information from the EPD / FSD, there were no DG / chemical wastes / fire incident / spillage records for the site.	60750817/R39.2/804	N/A	Aerial Photo 1.03 – 10.03	Yes	• Metals (Cr, Cu, Pb, Mn, Ni & Zn) • VOCs (Acetone, BTEX, MTBE & Trichloroethene) • SVOCs (PAHs) • PCRs
NMH-ST-006	MTR Spur Line Emergency Access Point No.5 (paved)	2,530 (Area within Project Site: 561)	• The site is located to the north of Pak Shek Au Village and was inaccessible with gate closed for detailed site walkover at the time of reporting. • Based on peripheral observations and the drone survey, the site was MTR Spur Line Emergency Access Point No.5. • Based on information from the EPD / FSD, there were no DG / chemical wastes / spillage records for the site. • A fire incident (FSD130) was record in the building. Excessive heat generated from a street hydrant pump ignited its insulating materials and in turn set fire to the same. The fire was put out with carbon dioxide fire extinguisher. Potential land contamination corresponding to this incident is not anticipated.	60750817/R39.2/805	No past potentially contaminating activities were identified from historical aerial photo. Previous EIA Report (AEIAR-175/2013) identified the site (KTN-87a) as MTR Spur Line Emergency Access Point No.5 with "B - Low Potential" on land contamination and recommended for revisit.	Aerial Photo 1.02 – 10.02	No	Not Applicable
NMH-ST-007	Open storage for construction equipment (paved)	1,128 (Area within Project Site: 738)	• The site is located to the north of Pak Shek Au Village and was inaccessible for detailed site walkover at the time of reporting. • Based on peripheral observations and the drone survey, construction equipment (excavators) storage was observed. • Based on information from the EPD / FSD, there were no DG / chemical wastes / fire incident / spillage records for the site.	60750817/R39.2/806	No past potentially contaminating activities were identified from historical aerial photo. Previous EIA Report (AEIAR-175/2013) identified the site (KTN-87b) as workshop or storage with "B - Low Potential" on land contamination and recommended for revisit. Previous EIA Report (EIA AEIAR-259/2024) identified the site as potentially contaminated site of vehicle maintenance (PAA002) and warehouse (PAA003). The sites were inaccessible and no drillhole was proposed. Site re-appraisal was recommended.	Aerial Photo 1.02 – 10.02	Yes	• Metals (Full List) • VOCs (Full List) • SVOCs (Full List) • PCRs

Site ID (Affected Lot No. / Address)[1]	Suspected Current Land Use	Approx. Site Area (m^2)	Site Observation and Information from EPD/FSD	Corresponding Photographic Record Reference in Appendix 7.5	Other Past Potentially Contaminating Land Uses / Activities	Corresponding Historical Aerial Photograph Reference in Appendix 7.2	Potentially Contaminated Site?	Potential COCs[3]
NMH-ST-008	Heavy vehicle maintenance / repair workshop (paved)	1,610	• The site is located to the north of Pak Shek Au Village and was inaccessible for detailed site walkover at the time of reporting. • Based on peripheral observations and the drone survey, vehicle workshop and some dump trucks were observed. • Based on information from the EPD / FSD, there were no DG / chemical wastes / fire incident / spillage records for the site.	60750817/R39.2/806	No past potentially contaminating activities were identified from historical aerial photo. Previous EIA Report (AEIAR-175/2013) identified the site (KTN-87c) as pig-oil manufacturer with "B - Low Potential" on land contamination and recommended for revisit. Previous EIA Report (AEIAR-259/2024) identified the site as potentially contaminated site of vehicle maintenance (PAA002 & PAA004). Further site investigation was recommended at PAA002. PAA004 was inaccessible and no drillhole was proposed. Site re-appraisal was recommended.	Aerial Photo 1.02 – 10.02	Yes	• Metals (Cr, Cu, Pb, Mn, Ni & Zn) • VOCs (Acetone, BTEX, MTBE & Trichloroethene) • SVOCs (PAHs) • PCRs
NMH-ST-009 (DD96 Lot 621)	Open storage (paved)	3,367	• The site is located to the west of Pak Shek Au Village and was inaccessible for detailed site walkover at the time of reporting. • Based on peripheral observations and the drone survey, open storage with containers were observed. • Based on information from the EPD / FSD, there were no DG / fire incident / spillage records for the site. There was 1 invalid CWP record registered for waste material recycling for the site.	60750817/R39.2/807	No past potentially contaminating activities were identified from historical aerial photo. The CWP record reflected previous recycling operation. Previous EIA Report (EIA AEIAR-259/2024) identified the site as open carpark and office (PAA006). No potential contamination source was identified. Site re-appraisal was recommended.	Aerial Photo 1.02 – 10.02	Yes	• Metals (Full List) • VOCs (Full List) • SVOCs (Full List) • PCRs • PCBs
NMH-ST-010 (DD96 Lot 232, 624)	Logistic company (paved)	3,293	• The site is located to the west of Pak Shek Au Village and was inaccessible for detailed site walkover at the time of reporting. • Based on peripheral observations and the drone survey, warehouses with trucks were observed at the site. • Based on information from the EPD / FSD, there were no DG / fire incident / spillage records for the site. There was 1 valid CWP record registered for engineering contractors and engineering merchants for the site. Potential equipment maintenance may be undertaken.	60750817/R39.2/807	No past potentially contaminating activities were identified from historical aerial photo. Previous EIA Report (EIA AEIAR-259/2024) identified the site as potentially contaminated site of open carpark (previous workshop) (PAA005). Further site investigation was recommended.	Aerial Photo 1.02 – 10.02	Yes	• Metals (Cr, Cu, Pb, Mn, Ni & Zn) • VOCs (Acetone, BTEX, MTBE & Trichloroethene) • SVOCs (PAHs) • PCRs

Site ID (Affected Lot No. / Address)[1]	Suspected Current Land Use	Approx. Site Area (m^2)	Site Observation and Information from EPD/FSD	Corresponding Photographic Record Reference in Appendix 7.5	Other Past Potentially Contaminating Land Uses / Activities	Corresponding Historical Aerial Photograph Reference in Appendix 7.2	Potentially Contaminated Site?	Potential COCs[3]
NMH-ST-011 (DD96 Lot 497 RP)	Scrapyard	1,146 (Area within Project Site: 987)	• The site is located near Chau Tau Tsuen and was inaccessible for detailed site walkover at the time of reporting. • Based on peripheral observations and the drone survey, the site was a scrapyard for metal recycling. • Based on information from the EPD / FSD, there were no DG / fire incident / spillage records for the site. There was 1 valid CWP record registered for recycling for the site.	60750817/R39.2/808	No past potentially contaminating activities were identified from historical aerial photo. Previous EIA Report (EIA AEIAR-261/2024) identified the site as potentially contaminated site of Scrapyard (E055). Further site investigation was recommended.	Aerial Photo 1.02 – 10.02	Yes	• Metals • VOCs (Acetone, BTEX, MTBE & Trichloroethene) • SVOCs (PAHs) • PCBs • PCRs
NMH-ST-012	Vacant land	979	• The site is located near Chau Tau Tsuen. The site was vacant and fenced. • Based on information from the EPD / FSD, there were no DG / chemical wastes / fire incident / spillage records for the site.	60750817/R39.2/808	No past potentially contaminating activities were identified from historical aerial photo. Previous EIA Report (AEIAR-261/2024) identified the site (E056) as vacant land and did not recommend further investigation.	Aerial Photo 1.02 – 10.02	No	Not Applicable
NMH-ST-013	Industrial land use[2] - Electricity Substation	48	• The site is located near Chau Tau Tsuen and was inaccessible for detailed site walkover at the time of reporting. • The site was Chau Tau Substation. • Based on information from the EPD / FSD, there were no DG / chemical wastes / fire incident / spillage records for the site.	60750817/R39.2/808	No past potentially contaminating activities were identified from historical aerial photo. Previous EIA Report (AEIAR-261/2024) identified the site (E057) as electricity substation and recommended for revisit.	Aerial Photo 1.02 – 10.02	TBC[2]	TBC
NMH-ST-014 (DD98 Lot 742 RP)	Vehicle workshop (paved)	1,310 (Area within Project Site: 95)	• The site is located at the south of Kwu Tung Road and was inaccessible for detailed site walkover at the time of reporting. • Based on peripheral observations and the drone survey, the site appeared to be vehicle workshop with container storage • Based on information from the EPD / FSD, there were no DG / chemical wastes / fire incident / spillage records for the site.	60750817/R39.2/809	No past potentially contaminating activities were identified from historical aerial photo. Previous EIA Report (EIA AEIAR-261/2024) identified the site as potentially contaminated site of vehicle workshop (E069). Further site investigation was recommended.	Aerial Photo 1.02 – 10.02	Yes	• Metals (Cr, Cu, Pb, Mn, Ni & Zn) • VOCs (Acetone, BTEX, MTBE & Trichloroethene) • SVOCs (PAHs) • PCRs
NMH-ST-015	Vehicle maintenance and repair workshop (paved)	317 (Area within Project Site: 315)	• The site is located at Pak Shek Au Village and was inaccessible for detailed site walkover at the time of reporting. • Based on peripheral observations and the drone survey, vehicle repair and maintenance activities were observed. The site was a vehicle maintenance and repair workshop. • Based on information from the EPD / FSD, there were no DG / chemical wastes / fire incident / spillage records for the site.	60750817/R39.2/810	No past potentially contaminating activities were identified from historical aerial photo. Previous EIA Report (AEIAR-175/2013) identified the site (KTN-86) as vehicle repair and maintenance (small scale) (paved) with "B - Low Potential" on land contamination and recommended for revisit.	Aerial Photo 1.03 – 10.03	Yes	• Metals (Cr, Cu, Pb, Mn, Ni & Zn) • VOCs (Acetone, BTEX, MTBE & Trichloroethene) • SVOCs (PAHs) • PCRs

Site ID (Affected Lot No. / Address)[1]	Suspected Current Land Use	Approx. Site Area (m^2)	Site Observation and Information from EPD/FSD	Corresponding Photographic Record Reference in Appendix 7.5	Other Past Potentially Contaminating Land Uses / Activities	Corresponding Historical Aerial Photograph Reference in Appendix 7.2	Potentially Contaminated Site?	Potential COCs[3]
NMH-ST-016	Industrial land use[2] - warehouse (paved)	67	• The site is located at the north of Castle Peak Road and was inaccessible for detailed site walkover at the time of reporting. • Based on peripheral observations and the drone survey, a suspected warehouse (temporary structure) was observed. • Based on information from the EPD / FSD, there were no DG / chemical wastes / fire incident / spillage records for the site.	60750817/R39.2/810	No past potentially contaminating activities were identified from historical aerial photo. Previous EIA Report (AEIAR-175/2013) identified the site (KTN-84a) as storage warehouse with "U - limited for access for visual inspection during the survey" on land contamination and recommended for revisit.	Aerial Photo 1.03 – 10.03	TBC[2]	TBC
NMH-ST-017	Construction site offices (paved)	5,509 (Area within Project Site: 2,255)	• The site is located at the north of Castle Peak Road and was inaccessible for detailed site walkover at the time of reporting. • Based on peripheral observations and the drone survey, the site was site offices for Contract No. ND/2019/01 Kwu Tung North New Development Area, Phase 1: Site Formation and Infrastructure Works • Based on information from the EPD / FSD, there were no DG / chemical wastes / fire incident / spillage records for the site.	60750817/R39.2/811	The site was a trailer park from 2002 to 2020. Previous EIA Report (AEIAR-175/2013) identified the site (KTN-84) as trailer park with repair and maintenance activities with "A - High Potential" on land contamination. Further site investigation is recommended.	Aerial Photo 1.03 – 10.03	Yes	• Metals (Cr, Cu, Pb, Mn, Ni & Zn) • VOCs (Acetone, BTEX, MTBE & Trichloroethene) • SVOCs (PAHs) • PCRs
NMH-ST-018 (DD95 FSSTL 277)	Kwu Tung North Multi-welfare Services Complex, Kwu Tung North Substation and Pak Sau Road (paved)	9,254 (Area within Project Site: 19)	• The site is located at the north of Castle Peak Road and was accessible for detailed site walkover at the time of reporting. • Based on peripheral observations and the drone survey, the site consisted of Kwu Tung North Multi-Welfare Services Complex, Kwu Tung North Substation and Pak Sau Road. • Based on information from the EPD / FSD, there were no DG / chemical wastes / fire incident / spillage records for the site.	60750817/R39.2/812	No past potentially contaminating activities were identified from historical aerial photo. Previous EIA Report (AEIAR-175/2013) identified the sites (KTN-82 and KTN-83) as vacant (partially paved) with "B - Low Potential" on land contamination and recommended for revisit.	Aerial Photo 1.03 – 10.03	No	Not Applicable
NMH-ST-019 (DD98 Lot 710 RP)	Open storage (paved)	342 (Area within Project Site: 322)	• The site is located at the south of Kwu Tung Road and was inaccessible for detailed site walkover at the time of reporting. • Based on peripheral observations and drone survey, the site is a contractor storage. Storage of metal materials and an excavator were observed on site. • Based on information from the EPD / FSD, there were no DG / chemical wastes / fire incident / spillage records for the site.	60750817/R39.2/813	N/A	Aerial Photo 1.03 – 10.03	Yes	• Metals (Full List) • VOCs (Full List) • SVOCs (Full List) • PCRs

Site ID (Affected Lot No. / Address)[1]	Suspected Current Land Use	Approx. Site Area (m^2)	Site Observation and Information from EPD/FSD	Corresponding Photographic Record Reference in Appendix 7.5	Other Past Potentially Contaminating Land Uses / Activities	Corresponding Historical Aerial Photograph Reference in Appendix 7.2	Potentially Contaminated Site?	Potential COCs[3]
NMH-ST-020 (DD98 Lot 708, 709, 712 RP)	Vehicle workshop (paved)	843 (Area within Project Site: 321)	• The site is located at the south of Kwu Tung Road and was inaccessible for detailed site walkover at the time of reporting. • Based on peripheral observations and drone survey, suspected vehicle workshop was observed. • Based on information from the EPD / FSD, there were no DG / chemical wastes / fire incident / spillage records for the site.	60750817/R39.2/813	N/A	Aerial Photo 1.03 – 10.03	Yes	• Metals (Cr, Cu, Pb, Mn, Ni & Zn) • VOCs (Acetone, BTEX, MTBE & Trichloroethene) • SVOCs (PAHs) • PCRs
NMH-ST-021 (DD98 Lot 726, 727)	Industrial land use[2] - warehouses (paved)[2]	5,658	• The site is located at the south of Kwu Tung Road and was inaccessible for detailed site walkover at the time of reporting. • Based on peripheral observations and drone survey, some suspected warehouses and trucks were observed. • Based on information from the EPD / FSD, there were no DG / chemical wastes / fire incident / spillage records for the site.	60750817/R39.2/814	N/A	Aerial Photo 1.03 – 10.03	TBC[2]	TBC
NMH-ST-022	Car storage (paved)	281 (Area within Project Site: 22)	• The site is located near Ki Lun Tsuen and was inaccessible for detailed site walkover at the time of reporting. • Based on peripheral observations and drone survey, vehicle parking was observed. • Based on information from the EPD / FSD, there were no DG / chemical wastes / fire incident / spillage records for the site.	60750817/R39.2/815	No past potentially contaminating activities were identified from historical aerial photo. Previous EIA Report (AEIAR-261/2024) identified the site (S063) as a car park and did not recommend further investigation	Aerial Photo 1.06 – 10.06	No	Not Applicable
NMH-ST-023 (DD98 Lot 443, 445, 446, 447, 448, 450, 451, 452, 458 S.B ss.1, 461, 462, 463, 464, 465, 466, 467, 468 RP, 468 S.A, 469, 470, 471 RP, 471 S.A, 472, 473, 474, 475, 476, 477, 478, 479, 480, 481, 482, 483, 484)	Pig farm	12,965 (Area within Project Site: 7,324)	• The site is located near Ki Lun Tsuen and was inaccessible for detailed site walkover at the time of reporting. • Based on peripheral observations and drone survey, the site is a pig farm. • Based on information from the EPD / FSD, there were no DG / chemical wastes / fire incident / spillage records for the site.	60750817/R39.2/816	No past potentially contaminating activities were identified from historical aerial photo. Previous EIA Report (AEIAR-261/2024) identified the site (S066) as a pig farm and did not recommend further investigation	Aerial Photo 1.06 – 10.06	No	Not Applicable

Site ID (Affected Lot No. / Address)[1]	Suspected Current Land Use	Approx. Site Area (m^2)	Site Observation and Information from EPD/FSD	Corresponding Photographic Record Reference in Appendix 7.5	Other Past Potentially Contaminating Land Uses / Activities	Corresponding Historical Aerial Photograph Reference in Appendix 7.2	Potentially Contaminated Site?	Potential COCs[3]
NMH-ST-024 (DD98 Lot 455. 456, 457, 458 S.B RP, 531, 532, 533, 534)	Pig farm	6,824 (Area within Project Site: 4,036)	• The site is located near Ki Lun Tsuen and was inaccessible for detailed site walkover at the time of reporting. • Based on peripheral observations and drone survey, the site is a pig farm. • Based on information from the EPD / FSD, there were no DG / chemical wastes / fire incident / spillage records for the site.	60750817/R39.2/817	No past potentially contaminating activities were identified from historical aerial photo. Previous EIA Report (AEIAR-261/2024) identified the site (S065) as a pig farm and did not recommend further investigation	Aerial Photo 1.06 – 10.06	No	Not Applicable
NMH-ST-025 (DD104 Lot 38, 42)	Open storage (partially paved)	2,580 (Area within Project Site: 1,300)	• The site is located at the southeast area of Ngau Tam Mei and was inaccessible for detailed site walkover at the time of reporting. • Based on peripheral observations and drone survey, construction materials and equipment storage were observed. • Based on information from the EPD / FSD, there were no DG / chemical wastes / fire incident / spillage records for the site.	60750817/R39.2/818	No other past potentially contaminating activities were identified from historical aerial photo. Previous EIA Report (AEIAR-271/2025) identified the site as potentially contaminated site of construction material and equipment storage (S63). Further site investigation was recommended.	Aerial Photo 1.10 – 10.10	Yes	• Metals (Full List) • VOCs (Full List) • SVOCs (Full List) • PCRs
NMH-ST-026 (DD104 Lot 184, 196)	Open storage (unpaved)	491 (Area within Project Site: 295)	• The site is located at the southeast area of Ngau Tam Mei and was inaccessible for detailed site walkover at the time of reporting. • Based on peripheral observations and drone survey, the site of mostly vacant with some construction materials (bamboo, pipe, etc) observed. • Based on information from the EPD / FSD, there were no DG / chemical wastes / fire incident / spillage records for the site.	60750817/R39.2/818	No other past potentially contaminating activities were identified from historical aerial photo. Previous EIA Report (AEIAR-271/2025) identified the site as potentially contaminated site of construction material and equipment storage (S52). Further site investigation was recommended.	Aerial Photo 1.10 – 10.10	Yes	• Metals (Full List) • VOCs (Full List) • SVOCs (Full List) • PCRs
NMH-ST-027 (DD104 Lot 803, 804 S.A, 805 S.A, 806, 807, 808, 809, 810, 811, 812 RP, 4187 RP, 4187 S.C)	Open storage (paved)	6,966 (Area within Project Site: 330)	• The site is located at the south of Chuk Yau Road and was inaccessible for detailed site walkover at the time of reporting. • Based on peripheral observations and drone survey, the site was mostly vacant, with construction storage (trucks, excavators and construction materials) at the southeast corner. • Based on information from the EPD / FSD, there were no DG / chemical wastes / fire incident / spillage records for the site.	60750817/R39.2/819	The site was an open storage for cars in 2017. Previous EIA Report (AEIAR-271/2025) identified the site as potentially contaminated site of open storage of vehicles and construction materials (S43). Further site investigation was recommended.	Aerial Photo 1.12 – 10.12	Yes	• Metals (Full List) • VOCs (Full List) • SVOCs (Full List) • PCRs

Site ID (Affected Lot No. / Address)[1]	Suspected Current Land Use	Approx. Site Area (m^2)	Site Observation and Information from EPD/FSD	Corresponding Photographic Record Reference in Appendix 7.5	Other Past Potentially Contaminating Land Uses / Activities	Corresponding Historical Aerial Photograph Reference in Appendix 7.2	Potentially Contaminated Site?	Potential COCs[3]
NMH-ST-028 (DD104 Lot 3458 RP)	Vehicle workshop (paved)	283 (Area within Project Site: 263)	• The site is located at the east of San Tam Road and was inaccessible for detailed site walkover at the time of reporting. • Based on peripheral observations and drone survey, container storage and vehicle (chassis) were observed on site. • Based on information from the EPD / FSD, there were no DG / chemical wastes / fire incident / spillage records for the site.	60750817/R39.2/820	N/A	Aerial Photo 1.14 – 10.14	Yes	• Metals (Cr, Cu, Pb, Mn, Ni & Zn) • VOCs (Acetone, BTEX, MTBE & Trichloroethene) • SVOCs (PAHs) • PCRs
NMH-ST-029 (DD104 Lot 3482 RP, 3505, 3522, 3525, 3526)	Industrial land use[2] - warehouse (paved)[2]	649	• The site is located at the east of San Tam Road and was inaccessible for detailed site walkover at the time of reporting. • Based on peripheral observations and drone survey, suspected warehouses and vehicles were observed. • Based on information from the EPD / FSD, there were no DG / chemical wastes / fire incident / spillage records for the site.	60750817/R39.2/821	N/A	Aerial Photo 1.14 – 10.14	TBC[2]	TBC
NMH-ST-030 (DD104 Lot 3520 RP, 3521, 3522, 3525)	Industrial land use[2] - warehouse (paved)[2]	452	• The site is located at the east of San Tam Road and was inaccessible for detailed site walkover at the time of reporting. • Based on peripheral observations and drone survey, suspected warehouses and vehicles were observed. • Based on information from the EPD / FSD, there were no DG / chemical wastes / fire incident / spillage records for the site.	60750817/R39.2/821	N/A	Aerial Photo 1.14 – 10.14	TBC[2]	TBC
NMH-ST-031 (DD104 Lot 3513, 3841 S.B, 3842 S.A, 3843 S.A, 3847 S.A, 3874, 3875, 3876, 3877, 3878, 3883, 3884)	Warehouse (Storage of Pet Supplies and Gardening Goods)	3,102	• The site is located at the east of San Tam Road and was inaccessible for detailed site walkover at the time of reporting. • Based on peripheral observations and drone survey, warehouses and trucks were observed. • Based on information from the EPD / FSD, there were no DG / chemical wastes / fire incident / spillage records for the site.	60750817/R39.2/822	N/A	Aerial Photo 1.15 – 10.15	No	Not Applicable
NMH-ST-032 (DD104 Lot 3856 RP, 3871, 3872, 3873 RP, 3873 S.A, 3890, 3902, 3903 RP)	Industrial land use[2] - warehouse (paved)[2]	916	• The site is located at the east of San Tam Road and was inaccessible for detailed site walkover at the time of reporting. • Based on peripheral observations and drone survey, suspected warehouses and container storage were observed. • Based on information from the EPD / FSD, there were no DG / chemical wastes / fire incident / spillage records for the site.	60750817/R39.2/823	N/A	Aerial Photo 1.15 – 10.15	TBC[2]	TBC

Site ID (Affected Lot No. / Address)[1]	Suspected Current Land Use	Approx. Site Area (m^2)	Site Observation and Information from EPD/FSD	Corresponding Photographic Record Reference in Appendix 7.5	Other Past Potentially Contaminating Land Uses / Activities	Corresponding Historical Aerial Photograph Reference in Appendix 7.2	Potentially Contaminated Site?	Potential COCs[3]
NMH-ST-033 (DD104 Lot 4114)	Laboratory (PROVA Inspection & Testing Co. (HK) Limit) (paved)[2]	473	• The site is located at Mo Fan Heung and was inaccessible for detailed site walkover at the time of reporting. • Based on peripheral observations and drone survey, suspected warehouses and container storage were observed. • Prova Inspection & Testing Company (HK) Limited was a HOKLAS accredited laboratory (HOKLAS 141). It mainly conducted physical test for metal/welding. • Based on information from the EPD / FSD, there were no DG / chemical wastes / fire incident / spillage records for the site.	60750817/R39.2/824	N/A	Aerial Photo 1.15 – 10.15	No	Not Applicable
NMH-ST-034 (DD104 Lot 4115)	Open storage / Contractor warehouse (paved)[2]	697	• The site is located at Mo Fan Heung and was inaccessible for detailed site walkover at the time of reporting. • Based on peripheral observations and drone survey, the site was a contractor warehouse with some material storage at the open area. • Based on information from the EPD / FSD, there were no DG / chemical wastes / fire incident / spillage records for the site.	60750817/R39.2/824	N/A	Aerial Photo 1.15 – 10.15	No	Not Applicable
NMH-ST-035 (DD104 Lot 4123, 4124, 4125)	Carwash workshop (paved)	64	• The site is located at Mo Fan Heung and was inaccessible for detailed site walkover at the time of reporting. • Based on peripheral observations and drone survey, the site is a carwash workshop. The workshop provides car wash, applies paint protective film and car spraying. • Based on information from the EPD / FSD, there were no DG / chemical wastes / fire incident / spillage records for the site.	60750817/R39.2/824	N/A	Aerial Photo 1.15 – 10.15	Yes	• Metals (Cr, Cu, Pb, Mn, Ni & Zn) • VOCs (Acetone, BTEX, MTBE & Trichloroethene) • SVOCs (PAHs) • PCRs

Site ID (Affected Lot No. / Address)[1]	Suspected Current Land Use	Approx. Site Area (m^2)	Site Observation and Information from EPD/FSD	Corresponding Photographic Record Reference in Appendix 7.5	Other Past Potentially Contaminating Land Uses / Activities	Corresponding Historical Aerial Photograph Reference in Appendix 7.2	Potentially Contaminated Site?	Potential COCs[3]
NMH-ST-036 (DD107 Lot 94, 95 S.A, 95 S.C, 96 S.A, 96 S.B, 98 S.A, 98 S.B, 98 S.C, 115 S.A, 115 S.B, 115 S.C, 115 S.D, 116 S.A, 116 S.C, 116 S.D, 116 S.F, 169 S.B, 169 S.C, 169 S.D, 169 S.E, 169 S.F, 170 S.A, 170 S.B, 171 S.A, 171 S.B, 172 S.A, 172 S.B, 173 S.A, 173 S.B, 173 S.C, 174 S.A, 174 S.B)	Open storage (paved)	1,576 (Area within Project Site: 1,297)	• The site is located at Mo Fan Heung and was inaccessible for detailed site walkover at the time of reporting. • Based on peripheral observations and drone survey, the site was a metal scaffold company. Storage of metal scaffold materials and scissor lifts (electric) were observed. • Based on information from the EPD / FSD, there were no DG / chemical wastes / fire incident / spillage records for the site.	60750817/R39.2/825	N/A	Aerial Photo 1.15 – 10.15	No	Not Applicable
NMH-ST-037 (DD107 Lot 56 S.D RP, 58 S.A, 58 S.B, 59 S.B, 59 S.C, 207 S.A, 208 S.A, 209 S.B, 209 S.C)	Open storage (unpaved)	752	• The site is located at Mo Fan Heung and was inaccessible for detailed site walkover at the time of reporting. • Based on peripheral observations and drone survey, construction material storage (wooden formworks and metal scaffold planks) container storage was observed. • Based on information from the EPD / FSD, there were no DG / chemical wastes / fire incident / spillage records for the site.	60750817/R39.2/826	N/A	Aerial Photo 1.15 – 10.15	No	Not Applicable
NMH-ST-038 (DD107 Lot 5, 12, 13, 14 S.B ss.2, 14 S.C RP, 15, 16, 17, 18, 31 S.B RP, 31 S.B ss.3, 36 RP, 55 S.A)	Open storage / Contractor warehouse (paved)	2,737 (Area within Project Site: 613)	• The site is located at the west of Castle Peak Road - Tam Mei and was inaccessible for detailed site walkover at the time of reporting. • Based on peripheral observations and drone survey, the site is open storage for construction equipment. • Based on information from the EPD / FSD, there were no DG / chemical wastes / fire incident / spillage records for the site.	60750817/R39.2/827	N/A	Aerial Photo 1.15 – 10.15	Yes	• Metals (Full List) • VOCs (Full List) • SVOCs (Full List) • PCRs
NMH-ST-039 (DD107 Lot 49 S.B RP, 59 S.B, 51 RP, 379 S.B, 381 RP, 383, 385 RP, 392 S.B, 394 S.A RP, 394 S.C, 395, 396, 397 RP, 398, 399 RP, 401 RP)	Open storage (paved)	6,127 (Area within Project Site: 5,902)	• The site is located at the north of Fung Kat Heung Road and was inaccessible with gate closed for detailed site walkover at the time of reporting. • Based on peripheral observations and drone survey, the site was a metal scaffold company. Storage of metal scaffold materials and container storage were observed. • Based on information from the EPD / FSD, there were no DG / chemical wastes / fire incident / spillage records for the site.	60750817/R39.2/828	No past potentially contaminating activities were identified from historical aerial photo. Previous EIA Report (EIA AEIAR-259/2024) identified the site as potentially contaminated site of warehouse (AUT012). The site was inaccessible and no drillhole was proposed. Site re-appraisal was recommended.	Aerial Photo 1.16 – 10.16	No	Not Applicable

Site ID (Affected Lot No. / Address)[1]	Suspected Current Land Use	Approx. Site Area (m^2)	Site Observation and Information from EPD/FSD	Corresponding Photographic Record Reference in Appendix 7.5	Other Past Potentially Contaminating Land Uses / Activities	Corresponding Historical Aerial Photograph Reference in Appendix 7.2	Potentially Contaminated Site?	Potential COCs[3]
NMH-ST-040 (DD107 Lot 51 S.A, 56 S.C, 57 S.A ss.1, 57 S.B ss.1)	Industrial land use[2] - Warehouse (paved)[2]	351	• The site is located at Mo Fan Heung and was inaccessible for detailed site walkover at the time of reporting. • Based on peripheral observations and drone survey, warehouses and lorry parking were observed. • Based on information from the EPD / FSD, there were no DG / chemical wastes / fire incident / spillage records for the site.	60750817/R39.2/829	N/A	Aerial Photo 1.16 – 10.16	TBC[2]	TBC
NMH-ST-041 (DD107 Lot 51 S.A, 57 S.B, 57 S.B ss.1, 211 S.A, 211 S.B, 212 S.A, 212 S.B, 213)	Vehicle maintenance and repair workshop (paved)	526	• The site is located at Mo Fan Heung and was inaccessible for detailed site walkover at the time of reporting. • Based on peripheral observations and drone survey, the site was a vehicle workshop. Maintenance and repair activities were observed. • Based on information from the EPD / FSD, there were no DG / chemical wastes / fire incident / spillage records for the site.	60750817/R39.2/829	N/A	Aerial Photo 1.16 – 10.16	Yes	• Metals (Cr, Cu, Pb, Mn, Ni & Zn) • VOCs (Acetone, BTEX, MTBE & Trichloroethene) • SVOCs (PAHs) • PCRs
NMH-ST-042 (DD107 Lot 50 S.A, 50 S.B, 57 S.B ss.1, 211 S.A, 211 S.B, 213, 215 S.C, 216 S.A, 216 S.B, 217 S.A, 217 S.B, 217 S.C)	Warehouse (mini-storages) (paved)	1,173 (Area within Project Site: 1,147)	• The site is located at Mo Fan Heung and was inaccessible for detailed site walkover at the time of reporting. • Based on peripheral observations and drone survey, the site was a warehouse (Hongkong Storage). Hongkong Storage was a storage company for household (mini storage) and document storage. • Based on information from the EPD / FSD, there were no DG / chemical wastes / fire incident / spillage records for the site.	60750817/R39.2/829	N/A	Aerial Photo 1.16 – 10.16	No	Not Applicable
NMH-ST-043 (DD107 Lot 192 S.A, 192 S.B, 210 S.A, 214 S.A, 215 S.A, 215 S.B, 265, 268 S.B, 269 S.A, 270 S.A, 270 S.B, 271, 272 S.A, 273 S.A, 274 S.A&B, 275, 276 S.A, 276 S.B, 277 S.A, 278, 279 S.A)	Industrial land use[2] - Warehouse (paved)[2]	2,130 (Area within Project Site: 1,229)	• The site is located at Mo Fan Heung and was inaccessible for detailed site walkover at the time of reporting. • Based on peripheral observations and the drone survey, warehouses were observed. • Based on information from the EPD / FSD, there were no DG / fire incident / spillage records for the site. • There was 1 valid CWP record registered for warehouse for the site. With reference to **Section 7.5.6.3**, the CWP is a home care product company.	60750817/R39.2/830	N/A	Aerial Photo 1.16 – 10.16	TBC[2]	TBC

Site ID (Affected Lot No. / Address)[1]	Suspected Current Land Use	Approx. Site Area (m^2)	Site Observation and Information from EPD/FSD	Corresponding Photographic Record Reference in Appendix 7.5	Other Past Potentially Contaminating Land Uses / Activities	Corresponding Historical Aerial Photograph Reference in Appendix 7.2	Potentially Contaminated Site?	Potential COCs[3]
NMH-ST-044 (DD107 Lot 49 S.A RP, 49 S.A ss.1, 49 S.B RP, 50 S.C, 218 S.A ss.1, 218 S.B, 219 S.A ss.1, 219 S.B, 220 S.A ss.1 RP, 220 S.A ss.1 S.A, 220 S.B)	Vehicle maintenance / repair workshop (paved)	525 (Area within Project Site: 196)	• The site is located at the east of San Tam Road and was inaccessible for detailed site walkover at the time of reporting. • Based on peripheral observations and the drone survey, the site was a vehicle maintenance / repair workshop. Engine oil storage was observed. • Based on information from the EPD / FSD, there were no DG / fire incident / spillage records for the site. There was 1 valid CWP record registered for motor vehicle repairing for the site.	60750817/R39.2/831	N/A	Aerial Photo 1.16 – 10.16	Yes	• Metals (Cr, Cu, Pb, Mn, Ni & Zn) • VOCs (Acetone, BTEX, MTBE & Trichloroethene) • SVOCs (PAHs) • PCRs
NMH-ST-045 (DD107 Lot 379 SB, 381 RP, 383)	Vehicle maintenance / repair workshop (paved)	278 (Area within Project Site: 47)	• The site is located at the north of Fung Kat Heung Road and was inaccessible for detailed site walkover at the time of reporting. • Based on peripheral observations and the drone survey, the site was a vehicle workshop. • Based on information from the EPD / FSD, there were no DG / chemical wastes / fire incident / spillage records for the site.	60750817/R39.2/832	No past potentially contaminating activities were identified from historical aerial photo. Previous EIA Report (AEIAR-259/2024) identified the site as potentially contaminated site of open storage and residential (AUT002). Further site investigation was recommended.	Aerial Photo 1.16 – 10.16	Yes	• Metals (Full List) • VOCs (Full List) • SVOCs (Full List) • PCRs
NMH-ST-046 (DD107 Lot 382, 418 RP, 419 S.A ss.1 RP, 420 S.B RP, 420 S.C RP)	Workshop (paved)	1,034 (Area within Project Site: 905)	• The site is located at the north of Fung Kat Heung Road and was inaccessible for detailed site walkover at the time of reporting. • Based on peripheral observations and the drone survey, workshop/warehouses were observed. The occupant company was a truck tailboard company (Sanda Engineering Limited). • Sign board on site also indicate the site was occupied by car demolition company (幫得手拆車環保回收有限公司). demolition/scrapping would have been undertaken. • Based on information from the EPD / FSD, there were no DG / fire incident / spillage records for the site. There was 1 valid CWP record registered for concrete mixing machinery maintenance and repair for the site.	60750817/R39.2/832	No past potentially contaminating activities were identified from historical aerial photo. Previous EIA Report (AEIAR-259/2024) identified the site as potentially contaminated site of workshop (AUT008). The site was inaccessible and no drillhole was proposed. Site re-appraisal was recommended.	Aerial Photo 1.16 – 10.16	Yes	• Metals (Cr, Cu, Pb, Mn, Ni & Zn) • VOCs (Acetone, BTEX, MTBE & Trichloroethene) • SVOCs (PAHs) • PCRs
NMH-ST-047 (DD107 Lot 380 S.C PR, 381 RP, 420 S.A, 420 S.A ss.1, 420 S.B RP, 420 S.C RP, 421, 422 S.B, 436)	Construction materials storage (paved)	1,051 (Area within Project Site: 140)	• The site is located at the south of Fung Kat Heung Road and was inaccessible for detailed site walkover at the time of reporting. • Based on peripheral observations and the drone survey, the site stored various construction materials for sale including sand, gravel, brick, bagged cement. • Based on information from the EPD / FSD, there were no DG / chemical wastes / fire incident / spillage records for the site.	60750817/R39.2/833	No past potentially contaminating activities were identified from historical aerial photo. Previous EIA Report (AEIAR-259/2024) identified the site as potentially contaminated site of open storage (AUT003). No potential contamination source was identified. Site re-appraisal was recommended.	Aerial Photo 1.16 – 10.16	No	Not Applicable

Site ID (Affected Lot No. / Address)[1]	Suspected Current Land Use	Approx. Site Area (m^2)	Site Observation and Information from EPD/FSD	Corresponding Photographic Record Reference in Appendix 7.5	Other Past Potentially Contaminating Land Uses / Activities	Corresponding Historical Aerial Photograph Reference in Appendix 7.2	Potentially Contaminated Site?	Potential COCs[3]
NMH-ST-048 (DD107 Lot 423 RP, 428 RP)	Industrial land use[2] - warehouses (paved)	1,603 (Area within Project Site: 1,362)	• The site is located at the south of Fung Kat Heung Road and was inaccessible for detailed site walkover at the time of reporting. • Based on peripheral observations and the drone survey, container storage and suspected warehouses were observed. • Based on information from the EPD / FSD, there were no DG / chemical wastes / fire incident / spillage records for the site.	60750817/R39.2/833	No past potentially contaminating activities were identified from historical aerial photo. Previous EIA Report (AEIAR-259/2024) identified the site as potentially contaminated site of warehouse (AUT004). The site was inaccessible and no drillhole was proposed. Site re-appraisal was recommended.	Aerial Photo 1.16 – 10.16	TBC [2]	TBC
NMH-ST-049 (DD107 Lot 422 S.A, 422 S.B, 428 RP, 429 RP, 430 S.B, 431, 434, 435, 436, 437 S.A, 437 S.B, 439)	warehouses (paved)	1,754 (Area within Project Site: 454)	• The site is located at the south of Fung Kat Heung Road and was inaccessible for detailed site walkover at the time of reporting. • Based on peripheral observations and the drone survey, suspected warehouses were observed. • Based on information from the EPD / FSD, there were no DG / chemical wastes / spillage records for the site. • A No. 1 Alarm Fire (FSD168) was record at Site NMH-ST-049. Given the nature of the incidents as No. 1 Alarm Fire, leakage/spillage of chemicals or dangerous goods are not anticipated. Potential land contamination issues associated with the incidents is not expected.	60750817/R39.2/833	No past potentially contaminating activities were identified from historical aerial photo. Previous EIA Report (AEIAR-259/2024) identified the site as potentially contaminated site of warehouse (AUT005). Further site investigation is recommended	Aerial Photo 1.16 – 10.16	Yes	• Metals (Full List) • VOCs (Full List) • SVOCs (Full List) • PCRs
NMH-ST-050 (DD107 Lot 1910 RP)	Sauce factory (paved)	27,938 (Area within Project Site: 2,227)	• The site is located at the west of Castle Peak Road - Tam Mei and was inaccessible for detailed site walkover at the time of reporting. • Based on peripheral observations and the drone survey, Koon Chun Hing Kee Soy & Sauce Factory Limited was observed. • Based on information from the FSD, there were 2 DG licenses issued (DG Record No.15 and 16 in **Table 7.4** refers) within the site. The 2 DG record are diesel storage, but the location of diesel storage was not identified in the site. • Based on information from the EPD / FSD, there were no fire incident / spillage records for the site. There were 2 valid CWP records registered for manufacturing of sauce products and construction for the site respectively.	60750817/R39.2/834	N/A	Aerial Photo 1.16 – 10.16	TBC [4]	TBC [4]
NMH-ST-051 (DD107 Lot 1743 S.C RP)	Container storage / Open storage (paved)	43,889 (Area within Project Site: 5,360)	• The site is located at the west of Castle Peak Road - Tam Mei and was inaccessible for detailed site walkover at the time of reporting. • Based on peripheral observations and the drone survey, container storage was observed. • Based on information from the EPD / FSD, there were no DG / fire incident / spillage records for the site. There was 1 valid CWP record registered for metal recycling for the site.	60750817/R39.2/835	N/A	Aerial Photo 1.16 – 10.16	Yes	• Metals (Full List) • VOCs (Full List) • SVOCs (Full List) • PCRs • PCBs

Site ID (Affected Lot No. / Address)[1]	Suspected Current Land Use	Approx. Site Area (m^2)	Site Observation and Information from EPD/FSD	Corresponding Photographic Record Reference in Appendix 7.5	Other Past Potentially Contaminating Land Uses / Activities	Corresponding Historical Aerial Photograph Reference in Appendix 7.2	Potentially Contaminated Site?	Potential COCs[3]
NMH-ST-052	Contractor site office (paved)	3,602	• The site is located at the east of Tsing Long Highway (Toll Road) and was inaccessible for detailed site walkover at the time of reporting. • Based on peripheral observations and the drone survey, contractor site offices (CEDD Contract No.: GE/2021/16 Landslip Prevention and Mitigation Programme, 2019, Package C, Landslip Prevention and Mitigation Works) were observed. • Based on information from the EPD / FSD, there were no DG / spillage records for the site. There were 2 valid CWP records registered for the site. The CWP was for the contractor site office.	60750817/R39.2/836	N/A	Aerial Photo 1.17 – 10.17	No	Not Applicable
NMH-ST-053 (DD103 Lot 1066, DD107 Lot 1928)	School / highway / soccer court / river	26,561 (Area within Project Site: 2,881)	• The site is located at the east of Tsing Long Highway and was inaccessible for detailed site walkover at the time of reporting. • Based on peripheral observations and the drone survey, Yeun Long Highway is located at the west portion of the site. The middle part of the site is Anchors Academy and the northeast part is Kam Tin River. • Based on information from the EPD / FSD, there were no DG / spillage records for the site. There was 1 valid and 2 invalid CWP records registered for construction of the existing school. • A rubbish fire incident (FSD157) was record on Ko Po Path near emergency access point of West Rail Line at Site NMH-ST-053. Given the nature of the incidents as rubbish fire, leakage/spillage of chemicals or dangerous goods are not anticipated. Potential land contamination issues associated with the incidents is not expected.	60750817/R39.2/837	Open storage of scrap yard and metal was observed in 1993.	Aerial Photo 1.18 – 10.18	Yes	• Metals (Full List) • VOCs (Full List) • SVOCs (Full List) • PCRs • PCBs
NMH-ST-054	Industrial land use[2] - warehouses (paved)	1,870 (Area within Project Site: 371)	• The site is located at the south of Chun Shin Road and was inaccessible for detailed site walkover at the time of reporting. • Based on peripheral observations and the drone survey, suspected warehouses were observed. • Based on information from the EPD / FSD, there were no DG / chemical wastes / fire incident / spillage records for the site.	60750817/R39.2/838	No past potentially contaminating activities were identified from historical aerial photo. Previous EIA Report (AEIAR-259/2024) identified the site as potentially contaminated site of warehouse (PAA001). The site was inaccessible and no drillhole was proposed. Site re-appraisal was recommended.	Aerial Photo 1.02 – 10.02	TBC [2]	TBC

Site ID (Affected Lot No. / Address)[1]	Suspected Current Land Use	Approx. Site Area (m^2)	Site Observation and Information from EPD/FSD	Corresponding Photographic Record Reference in Appendix 7.5	Other Past Potentially Contaminating Land Uses / Activities	Corresponding Historical Aerial Photograph Reference in Appendix 7.2	Potentially Contaminated Site?	Potential COCs[3]
NMH-ST-055 (DD104 Lot 3400, 3401, 3402, 3403, 3404, 3405, 3406, 3407, 3408, 3413, 3414)	Industrial land use[2] - warehouses (paved)	1,093	• The site is located at the east of San Tam Road and was inaccessible for detailed site walkover at the time of reporting. • Based on the drone survey, suspected warehouses were observed. The site was not accessible for peripheral inspection as the site was enclosed by private land. • Based on information from the EPD / FSD, there were no DG / chemical wastes / fire incident / spillage records for the site.	60750817/R39.2/839	No past potentially contaminating activities were identified from historical aerial photo. Previous EIA Report (AEIAR-259/2024) identified the site as potentially contaminated site of warehouse (LHA008). The site was inaccessible and no drillhole was proposed. Site re-appraisal was recommended.	Aerial Photo 1.14 – 10.14	TBC [2]	TBC
NMH-ST-056 (DD104 Lot 3400)	warehouses (paved)	153	• The site is located at the east of San Tam Road and was inaccessible for detailed site walkover at the time of reporting. • Based on the drone survey, suspected warehouses were observed. The site was not accessible for peripheral inspection as the site was enclosed by private land. • Based on information from the EPD / FSD, there were no DG / chemical wastes / fire incident / spillage records for the site.	60750817/R39.2/839	No past potentially contaminating activities were identified from historical aerial photo. Previous EIA Report (AEIAR-259/2024) identified the site as potentially contaminated site of warehouse (LHA001). Further site investigation is recommended	Aerial Photo 1.14 – 10.14	Yes	• Metals (Full List) • VOCs (Full List) • SVOCs (Full List) • PCRs
NMH-ST-057	Heavy equipment storage (paved)	231	• The site is located at Pak Shek Au Village and was inaccessible for detailed site walkover at the time of reporting. • Based on the peripheral observations and drone survey, some excavators and warehouse were observed. • Based on information from the EPD / FSD, there were no DG / chemical wastes / fire incident / spillage records for the site.	60750817/R39.2/840	N/A	Aerial Photo 1.02 – 10.02	Yes	• Metals (Cr, Cu, Pb, Mn, Ni & Zn) • VOCs (Acetone, BTEX, MTBE & Trichloroethene) • SVOCs (PAHs) • PCRs
NMH-ST-058	Canteen / offices of car sales company (paved)[2]	1,814 (Area within Project Site: 1,597)	• The site is located at the North of Castle Peak Road and was accessible for detailed site walkover at the time of reporting. • Based on the peripheral observations and drone survey, the site includes a canteen, offices of car sales company and car parking. No vehicle maintenance activities were observed. • Based on information from the EPD / FSD, there were no DG / chemical wastes / fire incident / spillage records for the site.	60750817/R39.2/841	N/A	Aerial Photo 1.03 – 10.03	No	Not Applicable

Site ID (Affected Lot No. / Address)[1]	Suspected Current Land Use	Approx. Site Area (m^2)	Site Observation and Information from EPD/FSD	Corresponding Photographic Record Reference in Appendix 7.5	Other Past Potentially Contaminating Land Uses / Activities	Corresponding Historical Aerial Photograph Reference in Appendix 7.2	Potentially Contaminated Site?	Potential COCs[3]
NMH-ST-059	Construction Site	1,061 (Area within Project Site: 461)	• The site is located at Pak Shek Au Village. The site was part of the construction site of Kwu Tung North development and inaccessible for detailed site walkover at the time of reporting. • Based on peripheral observations and the drone survey, the site was a part of construction site forming new road. • Based on information from the EPD / FSD, there were no DG / chemical wastes / fire incident / spillage records for the site.	60750817/R39.2/842	No past potentially contaminating activities were identified from historical aerial photo. Previous EIA (AEIAR-175/2013) identified the site (KTN-88b) as Tung Hing Metal Workshop / Storage with " B - Low Potential" on land contamination and recommended for revisit.	Aerial Photo 1.03 – 10.03	No	Not Applicable
NMH-ST-060	Construction Site	5,237 (Area within Project Site: 309)	• The site is located at Pak Shek Au Village. The site was part of the construction site of Kwu Tung North development and inaccessible for detailed site walkover at the time of reporting. • Based on peripheral observations and the drone survey, the site was a part of construction site forming new road. • Based on information from the EPD / FSD, there were no DG / chemical wastes / fire incident / spillage records for the site.	60750817/R39.2/843	No past potentially contaminating activities were identified from historical aerial photo. Previous EIA (AEIAR-175/2013) identified the site (KTN-85) as abandoned pig farm with " B - Low Potential" on land contamination and recommended for revisit.	Aerial Photo 1.03 – 10.03	No	Not Applicable
NMH-ST-061	Industrial land use[2] - Warehouse (paved)	850	• The site is located near Ki Lun Tsuen and near Saddle Pass and was inaccessible for detailed site walkover at the time of reporting. • Based on peripheral observations and drone survey, a suspected warehouse and forklifts were observed. • Based on information from the EPD / FSD, there were no DG / chemical wastes / fire incident / spillage records for the site.	60750817/R39.2/844	N/A	Aerial Photo 1.06 – 10.06	TBC[2]	TBC
NMH-ST-062	Pig farm with village house (paved)	1,557	• The site is located near Ki Lun Tsuen and near Saddle Pass and was inaccessible for detailed site walkover at the time of reporting. • Based on peripheral observations and drone survey, some farms and village houses were observed. • Based on information from the EPD / FSD, there were no DG / chemical wastes / fire incident / spillage records for the site.	60750817/R39.2/845	No past potentially contaminating activities were identified from historical aerial photo. Previous EIA (AEIAR-261/2024) identified the site (S064) as pig farm and did not recommend further investigation	Aerial Photo 1.06 – 10.06	No	Not Applicable
NMH-ST-063	Open storage for construction (paved)	6,786 (Area within Project Site: 2,037)	• The site is located at the south of Chuk Yau Road and was inaccessible for detailed site walkover at the time of reporting. • Based on peripheral observations and the drone survey, a suspected construction site depot was observed. Site offices and construction open storage was observed. • Based on information from the EPD / FSD, there were no DG / chemical wastes / fire incident / spillage records for the site.	60750817/R39.2/846	The site was an open storage for cars in 2017. Previous EIA Report (AEIAR-271/2025) identified the site as potentially contaminated site of a suspected construction site depot (S47). Further site investigation was recommended.	Aerial Photo 1.12 – 10.12	Yes	• Metals (Full List) • VOCs (Full List) • SVOCs (Full List) • PCRs

Site ID (Affected Lot No. / Address)[1]	Suspected Current Land Use	Approx. Site Area (m^2)	Site Observation and Information from EPD/FSD	Corresponding Photographic Record Reference in Appendix 7.5	Other Past Potentially Contaminating Land Uses / Activities	Corresponding Historical Aerial Photograph Reference in Appendix 7.2	Potentially Contaminated Site?	Potential COCs[3]
NMH-ST-064	Vehicle workshop and some vacant site (paved)	5,170 (Area within Project Site: 2,699)	• The site is located at the south of Chuk Yau Road and was inaccessible for detailed site walkover at the time of reporting. • Based on peripheral observations and the drone survey, a vehicle maintenance workshop and vacant area was observed. • Based on information from the EPD / FSD, there were no DG / chemical wastes / fire incident / spillage records for the site.	60750817/R39.2/847	The site was an open storage for cars in 2017. Previous EIA Report (AEIAR-271/2025) identified the site as potentially contaminated site of a vehicle maintenance workshop and open storage of vehicles (S45). Further site investigation was recommended.	Aerial Photo 1.12 – 10.12	Yes	• Metals (Full List) • VOCs (Full List) • SVOCs (Full List) • PCRs
NMH-ST-065	Industrial land use[2] - Village house with materials storage (paved)[2]	5,252 (Area within Project Site: 2,628)	• The site is located at the south of Chuk Yau Road and was inaccessible for detailed site walkover at the time of reporting. • Based on peripheral observations and drone survey, village houses with some materials storage were observed. • Based on information from the EPD / FSD, there were no DG / chemical wastes / fire incident / spillage records for the site.	60750817/R39.2/848	No past potentially contaminating activities were identified from historical aerial photo. Previous EIA Report (AEIAR-271/2025) identified the site as potentially contaminated site of temporary structures and suspected construction materials (S42). Further site investigation was recommended.	Aerial Photo 1.13 – 10.13	TBC[2]	TBC
NMH-ST-066	Farm (paved)	2,490 (Area within Project Site: 2,254)	• The site is located at the east of Ching Yau Road and was inaccessible for detailed site walkover at the time of reporting. • Based on peripheral observations and drone survey, farm with some temporary structures were observed. • Based on information from the EPD / FSD, there were no DG / chemical wastes / fire incident / spillage records for the site.	60750817/R39.2/849	Previous EIA Report (AEIAR-259/2024) identified the site as potentially contaminated site of open storage (NTM027). The site was inaccessible and no drillhole was proposed. Site re-appraisal was recommended. Further site appraisal and aerial photo review confirmed the site (NTM027) had been a farm since 1990s. Land contamination is not anticipated.	Aerial Photo 1.13 – 10.13	No	Not Applicable
NMH-ST-067	open storage (vehicle) and vehicle workshop (paved)	3,122 (Area within Project Site: 601)	• The site is located at the east of Ching Yau Road and was inaccessible for detailed site walkover at the time of reporting. • Based on peripheral observations and drone survey, the site was open storage (vehicle) and vehicle workshop. Vehicle, vehicle parts and container storage were observed. • Based on information from the EPD / FSD, there were no DG / chemical wastes / fire incident / spillage records for the site.	60750817/R39.2/849	No past potentially contaminating activities were identified from historical aerial photo. Previous EIA Report (AEIAR-259/2024) identified the site as potentially contaminated site of open storage (NTM026). The site was inaccessible and no drillhole was proposed. Site re-appraisal was recommended..	Aerial Photo 1.13 – 10.13	Yes	• Metals (Full List) • VOCs (Full List) • SVOCs (Full List) • PCRs

Site ID (Affected Lot No. / Address)[1]	Suspected Current Land Use	Approx. Site Area (m^2)	Site Observation and Information from EPD/FSD	Corresponding Photographic Record Reference in Appendix 7.5	Other Past Potentially Contaminating Land Uses / Activities	Corresponding Historical Aerial Photograph Reference in Appendix 7.2	Potentially Contaminated Site?	Potential COCs[3]
NMH-ST-068	Recycle company (Waste Disposal & Collect Center Limited) (paved)	2,023 (Area within Project Site: 325)	• The site is located at the west of Ching Yau Road and was inaccessible for detailed site walkover at the time of reporting. • Based on peripheral observations and drone survey, warehouses and container storage were observed. • Based on information from the EPD / FSD, there were no DG / chemical wastes / fire incident / spillage records for the site.	60750817/R39.2/850	Previous EIA Report (AEIAR-259/2024) identified the site as potentially contaminated site of recycling facilities (NTM011). Further site investigation was recommended.	Aerial Photo 1.13 – 10.13	Yes	• Metals • VOCs (Acetone, BTEX, MTBE & Trichloroethene) • SVOCs (PAHs) • PCBs • PCRs

Notes:

1 Refer to **Appendix 7.5** for locations of the sites. Given most of the sites are inaccessible, the boundaries shown are indicative and subject to confirmation in the further site appraisal.

2 TBC: To be confirmed. The actual land uses / activities within the suspected warehouse or industrial land uses could not be identified based on findings of peripheral site inspection, drone survey and / or review of historical aerial photographs. For warehouses, the presence and degree of contamination would greatly depend on the types of goods stored. For example, warehouses that stored general household goods (e.g. furniture and toys) are unlikely to cause contamination to the underlying soil and groundwater. For industrial land uses, the presence and degree of contamination would greatly depend on the actual site activity / operation. Further site appraisal should be carried out for these sites to confirm the presence of any land contamination issues.

3 The potential COCs were determined based on findings of the site appraisals and the Practice Guide and took into account current and historical land uses.
- Full list refers to the parameters as shown in Table 2.1 – Risk-Based Remediation Goals (RBRGs) for Soil & Soil Saturation Limit and Table 2.2 – RBRGs for Groundwater and Solubility Limit under Volatile Organic Compounds (VOCs), Semi-Volatile Organic Compounds (SVOCs), metals and Petroleum Carbon Ranges (PCRs) in the Guidance Manual for Use of Risk-Based Remediation Goals for Contaminated Land Management.
- BTEX includes benzene, toluene, ethylbenzene and total xylenes; MTBE refers to Methyl tert-Butyl Ether; PCBs refers to Polychlorinated biphenyls.
- PAHs include acenaphthene, acenaphthylene, anthracene, benzo(a)anthracene, benzo(a)pyrene, benzo(b)fluoranthene, benzo(g,h,i)perylene, benzo(k)fluoranthene, chrysene, dibenzo(a,h)anthracene, fluoranthene, fluorene, indeno(1,2,3-cd)pyrene, naphthalene, phenanthrene and pyrene.
-TBC – The COCs will be confirmed after the site is accessible for further site appraisal.

4 The location of diesel storage was not identified. Further site investigation will be required if the diesel storage fall within Project Boundary.

7.6 Site Investigation Plan

7.6.1.1 Based on the findings of the site appraisals, a total of 30 potentially contaminated sites and 15 sites suspected to be used for industrial purposes were identified within the Project Site. The remaining 23 sites are not considered as potentially contaminated sites. However, the potentially contaminated sites were still in operation and inaccessible at the time of reporting to determine the sampling locations for SI works. Similarly, the sites suspected to be used for industrial purposes were still in operation and inaccessible to confirm the existing land uses / activities and presence of any land contamination issues. Given that these concerned sites are still in operation and inaccessible at this stage in time and that land resumption and construction works under the Project would only commence after few years, further site appraisal should be carried out for the whole Project Site at later stage (i.e. after land resumption) during which sites are accessible to confirm the presence of any land contamination issues and determine the actual sampling and testing requirements for the concerned sites (refer to **Section 7.8** below for details).

7.6.1.2 For Site NMH-ST-050, the location of diesel storages was not identified. Further site investigation will be required if the diesel storage fall within Project Boundary.

7.6.1.3 A preliminary sampling and testing plan for the potentially contaminated sites was prepared based on the findings of site appraisals and with reference to EPD's Practice Guide. Details of the preliminary sampling and testing plan are shown in **Table 7.7**. Grid sampling strategy, in accordance with Table 2.1 of the Practice Guide, was adopted for the potentially contaminated sites. Subject to the findings of the detailed site walkover at later stage, extra sampling locations may be required for any additional potential sources of contamination (or 'hotspots') within the concerned sites. The key COCs for the potentially contaminated sites were determined with reference to EPD's Practice Guide and include volatile organic compounds (VOCs), semi-volatile organic compounds (SVOCs), metals, petroleum carbon ranges (PCRs) and PCBs.

Table 7.7 Preliminary Sampling and Testing Plan

Potentially Contaminated Site ID[1]	Current & Past Suspected Potentially Contamination Land Uses [2]	Approx. Site Area (m^2)	Approx. Area within Project Site (m^2)	Testing Parameters (Potential COCs)[3]	Recommended Minimum Number of Sampling Locations[4]
NMH-ST-005	Current: Vehicle maintenance workshop (paved) Past: N/A [5]	1,491	1,489	Metals (Cr, Cu, Pb, Mn, Ni & Zn), VOCs (Acetone, BTEX, MTBE & Trichloroethene), SVOCs (PAHs) and PCRs	12
NMH-ST-007	Current: Open storage for construction equipment (paved) Past: workshop or storage; vehicle maintenance and warehouse	1,128	738	Metals (Full List), VOCs (Full List), SVOCs (Full List) and PCRs	6
NMH-ST-008	Current: Heavy vehicle maintenance / repair workshop (unpaved) Past: pig-oil manufacturer; vehicle maintenance	1,610	1,610	Metals (Cr, Cu, Pb, Mn, Ni & Zn), VOCs (Acetone, BTEX, MTBE & Trichloroethene), SVOCs (PAHs) and PCRs	12
NMH-ST-009	Current: Open storage (paved) Past: Recycling; open carpark and office)	3,367	3,367	Metals (Full List), VOCs (Full List), SVOCs (Full List), PCRs and PCBs	17
NMH-ST-010	Current: Logistic company (paved) Past: open carpark (previous workshop)	3,293	3,293	Metals (Cr, Cu, Pb, Mn, Ni & Zn), VOCs (Acetone, BTEX, MTBE & Trichloroethene), SVOCs (PAHs) and PCRs	17
NMH-ST-011	Current: Scrapyard Past: Scrapyard	1,146	987	Metals, VOCs (Acetone, BTEX, MTBE & Trichloroethene), SVOCs (PAHs), PCBs and PCRs	6
NMH-ST-014	Current: Vehicle workshop (paved) Past: Vehicle workshop	1,310	95	Metals (Cr, Cu, Pb, Mn, Ni & Zn), VOCs (Acetone, BTEX, MTBE & Trichloroethene), SVOCs (PAHs) and PCRs	3
NMH-ST-015	Current: Vehicle maintenance and repair workshop (paved) Past: vehicle repair and maintenance (small scale) (paved)	317	315	Metals (Cr, Cu, Pb, Mn, Ni & Zn), VOCs (Acetone, BTEX, MTBE & Trichloroethene), SVOCs (PAHs) and PCRs	3

Potentially Contaminated Site ID[1]	Current & Past Suspected Potentially Contamination Land Uses [2]	Approx. Site Area (m^2)	Approx. Area within Project Site (m^2)	Testing Parameters (Potential COCs)[3]	Recommended Minimum Number of Sampling Locations[4]
NMH-ST-017	Current: Contractor site offices (paved) Past: Trailer park with repair and maintenance activities	5,509	2,255	Metals (Cr, Cu, Pb, Mn, Ni & Zn), VOCs (Acetone, BTEX, MTBE & Trichloroethene), SVOCs (PAHs) and PCRs	17
NMH-ST-019	Current: Open storage (paved) Past: N/A [5]	342	322	Metals (Full List), VOCs (Full List), SVOCs (Full List) and PCRs	3
NMH-ST-020	Current: Vehicle workshop (paved) Past: N/A [5]	843	321	Metals (Cr, Cu, Pb, Mn, Ni & Zn), VOCs (Acetone, BTEX, MTBE & Trichloroethene), SVOCs (PAHs) and PCRs	3
NMH-ST-025	Current: Open storage (partially paved) Past: Construction material and equipment storage	2,580	1,300	Metals (Full List), VOCs (Full List), SVOCs (Full List) and PCRs	12
NMH-ST-026	Current: Open storage (unpaved) Past: Construction material and equipment storage	491	295	Metals (Full List), VOCs (Full List), SVOCs (Full List) and PCRs	3
NMH-ST-027	Current: Open storage (unpaved) Past: Open storage of vehicles and construction materials	6,966	330	Metals (Full List), VOCs (Full List), SVOCs (Full List) and PCRs	3
NMH-ST-028	Current: Vehicle workshop (paved) Past: N/A [5]	283	263	Metals (Cr, Cu, Pb, Mn, Ni & Zn), VOCs (Acetone, BTEX, MTBE & Trichloroethene), SVOCs (PAHs) and PCRs	3
NMH-ST-035	Current: Carwash workshop (paved) Past: N/A [5]	64	64	Metals (Cr, Cu, Pb, Mn, Ni & Zn), VOCs (Acetone, BTEX, MTBE & Trichloroethene), SVOCs (PAHs) and PCRs	2
NMH-ST-038	Current: Open storage / Contractor warehouse (paved) Past: N/A [5]	2,737	613	Metals (Full List), VOCs (Full List), SVOCs (Full List) and PCRs	6
NMH-ST-041	Current: Vehicle maintenance and repair workshop (paved) Past: N/A [5]	526	526	Metals (Cr, Cu, Pb, Mn, Ni & Zn), VOCs (Acetone, BTEX, MTBE & Trichloroethene), SVOCs (PAHs) and PCRs	6

Potentially Contaminated Site ID[1]	Current & Past Suspected Potentially Contamination Land Uses [2]	Approx. Site Area (m^2)	Approx. Area within Project Site (m^2)	Testing Parameters (Potential COCs)[3]	Recommended Minimum Number of Sampling Locations[4]
NMH-ST-044	Current: Vehicle maintenance / repair workshop (paved) Past: N/A [5]	525	196	Metals (Cr, Cu, Pb, Mn, Ni & Zn), VOCs (Acetone, BTEX, MTBE & Trichloroethene), SVOCs (PAHs) and PCRs	3
NMH-ST-045	Current: Vehicle maintenance / repair workshop (paved) Past: Open storage and residential	278	47	Metals (Full List), VOCs (Full List), SVOCs (Full List) and PCRs	2
NMH-ST-046	Current: Workshop (paved) Past: Workshop	1,034	905	Metals (Cr, Cu, Pb, Mn, Ni & Zn), VOCs (Acetone, BTEX, MTBE & Trichloroethene), SVOCs (PAHs) and PCRs	6
NMH-ST-049	Current: Warehouses (paved) Past: Warehouses	1,754	454	Metals (Full List), VOCs (Full List), SVOCs (Full List) and PCRs	3
NMH-ST-051	Current: Container storage / Open area storage (paved) Past: N/A [5]	43,889	5,360	Metals (Full List), VOCs (Full List), SVOCs (Full List), PCRs and PCBs	28
NMH-ST-053	Current: No Past: Open storage of scrap yard and metal	26,561	2,881	Metals (Full List), VOCs (Full List), SVOCs (Full List), PCRs and PCBs	17
NMH-ST-056	Current: Suspected warehouses (paved) Past: Warehouse	153	153	Metals (Full List), VOCs (Full List), SVOCs (Full List) and PCRs	3
NMH-ST-057	Current: Heavy equipment storage (paved) Past: N/A [5]	231	231	Metals (Cr, Cu, Pb, Mn, Ni & Zn), VOCs (Acetone, BTEX, MTBE & Trichloroethene), SVOCs (PAHs) and PCRs	3
NMH-ST-063	Current: Open storage for construction (paved) Past: Suspected construction site depot	6,786	2,037	Metals (Full List), VOCs (Full List), SVOCs (Full List) and PCRs	17
NMH-ST-064	Current: Vehicle workshop and some vacant site (paved) Past: Vehicle maintenance workshop and open storage of vehicles	5,170	2,699	Metals (Full List), VOCs (Full List), SVOCs (Full List) and PCRs	17

Potentially Contaminated Site ID[1]	Current & Past Suspected Potentially Contamination Land Uses [2]	Approx. Site Area (m^2)	Approx. Area within Project Site (m^2)	Testing Parameters (Potential COCs)[3]	Recommended Minimum Number of Sampling Locations[4]
NMH-ST-067	Current: Suspected warehouses (paved) Past: Open storage	3,122	601	Metals (Full List), VOCs (Full List), SVOCs (Full List) and PCRs	6
NMH-ST-068	Current: Recycle company (Waste Disposal & Collect Center Limited) (paved) Past: Recycling facilities	2,023	325	Metals, VOCs (Acetone, BTEX, MTBE & Trichloroethene), SVOCs (PAHs), PCBs and PCRs	3

Notes:

(1) Refer to **Appendix 7.5** for locations of the identified potentially contaminated sites and the preliminary sampling plan.

(2) As the site was still in operation and inaccessible, only peripheral site inspections were carried out at the time of site walkover. The land use may be subject to change during further site appraisal when the site is accessible.

(3) The proposed testing parameters / COCs were determined based on findings of the site appraisals and with reference to the Practice Guide. The testing parameters / COCs will be confirmed after the site is accessible for re-appraisal.
- Full list refers to the parameters as shown in Table 2.1 – Risk-Based Remediation Goals (RBRGs) for Soil & Soil Saturation Limit and Table 2.2 – RBRGs for Groundwater and Solubility Limit under Volatile Organic Compounds (VOCs), Semi-Volatile Organic Compounds (SVOCs), metals and Petroleum Carbon Ranges (PCRs) in the Guidance Manual for Use of Risk-Based Remediation Goals for Contaminated Land Management.
- BTEX includes benzene, toluene, ethylbenzene and total xylenes; MTBE refers to Methyl tert-Butyl Ether.
- PAHs include acenaphthene, acenaphthylene, anthracene, benzo(a)anthracene, benzo(a)pyrene, benzo(b)fluoranthene, benzo(g,h,i)perylene, benzo(k)fluoranthene, chrysene, dibenzo(a,h)anthracene, fluoranthene, fluorene, indeno(1,2,3-cd)pyrene, naphthalene, phenanthrene and pyrene.
- Since RBRG value for metals except for Mercury, Benzo(a)anthracene, Benzo(a)pyrene, Benzo(g,h,i)perylene, Benzo(k)fluoranthene, bis-(2-Ethylhexyl)phthalate, Dibenzo(a,h)anthracene, Indeno(1,2,3-cd)pyrene and Phenol were not available for groundwater, the said parameters would not be tested in groundwater samples.

(4) With references to Table 2.1 of the Practice Guide. The actual number of sampling locations will be subject to the findings of the further site appraisal when the site is accessible.

(5) N/A indicates no other suspected potentially contamination land uses apart from the current land use.

7.7 Prediction and Evaluation of Environmental Impacts

7.7.1.1 Based on the findings of the site appraisals, a total of 30 potentially contaminated sites, 15 sites suspected to be used for industrial purposes and 23 sites with low land contamination potential were identified within the Project Site. However, all the concerned sites were still in operation or inaccessible for detailed site walkover or SI works.

7.7.1.2 Of the potentially contaminated sites identified in the site appraisals, some have been used for storage. The majority of these sites are usually kept for the storage of inert construction materials, whilst a small portion of the site is reserved for potentially contaminating activities such as vehicle / equipment maintenance area and the associated chemical handling / storage. In addition, apart from the sites currently used for storage, the remaining sites are relatively small in scale. As such, the contamination extent, if any, caused by the operations of the identified potentially contaminated sites is anticipated to be localised. Furthermore, the COCs identified included metals, VOCs, SVOCs and PCRs, which are readily treatable using established techniques and have been effectively remediated in Hong Kong using proven remediation techniques.

7.7.1.3 Given the above, the land contamination problems in the identified potentially contaminated sites would not be considered as insurmountable.

7.7.1.4 For sites suspected to be used as warehouses or other industrial purposes, the presence of land contamination within these sites would depend on their actual site activities. In particular for warehouses, the presence and degree of contamination would greatly depend on the types of goods stored, and whether contaminating activities such as vehicle maintenance / repairing, refuelling, storage of chemicals / chemical wastes etc., are / have been carried out. As such, further site appraisal should be carried out within these sites in next stage of the Project when site access is available to confirm the existing land uses / activities and to identify the presence of any potential contamination sources. In addition, as the sites are currently inaccessible or still in operation, and the construction works would not commence until 2027, the further site appraisal should also cover any further changes in the land uses within the Project Site since the time of this Report. If potentially contaminated sites are identified, intrusive SI and, if necessary, remediation works should also need to be carried out. The further site appraisal and, if necessary, intrusive SI and remediation works should be carried out prior to the commencement of construction works in accordance with EPD's Guidance Manual, Guidance Note and Practice Guide. Details and tentative programme of the further works are discussed in **Section 7.8** below.

7.7.1.5 The surveyed sites assessed to have low land contamination potential include grassland, residential areas, farms, contractor site offices and etc. Potential land contamination is not typically associated with activities undertaken within these types of facilities. Machinery/vehicle maintenance is not anticipated on these sites.

7.7.1.6 Any soil/groundwater contamination would be identified and properly remediated prior to the commencement of construction works if the further works as stated in **Section 7.8** are implemented. Land contamination impacts are considered surmountable to future occupants if the further works are followed and contaminated soil and groundwater (if any) are properly remediated according to

Remediation Action Plan(s) (RAP(s)) to be approved by EPD in the next stage of the Project.

7.8 Mitigation of Adverse Environmental Impacts

7.8.1 Recommended Further Works

7.8.1.1 Further site appraisal and, if necessary, intrusive SI and remediation works should be carried out in accordance with EPD's Guidance Manual, Guidance Note and Practice Guide.

7.8.1.2 Further site appraisal and submission of CAP(s) should be carried out for the whole Project Site at a later stage of the Project when site access is available and prior to the commencement of the SI works to confirm the existing land uses / activities, identify the presence of any potential contamination sources, and address any new contamination issues caused by the changes in site operation and/or land uses within the Project Site since the time of this Report. The associated SI works and any necessary remediation action are recommended to be carried out after the operation of concerned site(s) has ceased but prior to the commencement of construction works.

7.8.1.3 CAP(s), presenting findings of the further site appraisal, the latest site conditions and updated sampling strategy and testing protocol, should be submitted to EPD for endorsement. The SI works should be carried out according to EPD's approved CAP(s). Following the completion of SI works and receipt of laboratory test results, Contamination Assessment Report(s) (CAR(s)) should be submitted to EPD for approval to present the findings of the SI works and to discuss the presence, nature and extent of contamination. If contamination is identified, RAP(s) which provides details of the remedial actions for the identified contaminated soil and / or groundwater should also be submitted to EPD for approval.

7.8.1.4 With reference to the *Preliminary Feasibility Study on Developing the New Territories North*, the *Geochemical Atlas of Hong Kong* published by the Geotechnical Engineering Office of Civil Engineering and Development Department in 1999 and the Hong Kong Environmental Database (HKED) by EPD, natural occurrences of elevated level of arsenic exist over broad areas of the northern New Territories. A separate arsenic assessment and remediation plan should be prepared and submitted to EPD for endorsement prior to the commencement of construction works.

7.8.1.5 If necessary, remediation works should be conducted according to the approved RAP(s). Upon completion of remediation works, Remediation Report(s) (RR(s)) should be prepared and submitted to EPD to demonstrate that the decontamination work is adequate and is carried out in accordance with the approved CAR(s) and RAP(s). No construction works, other than decontamination works, at the respective identified contamination area (if any) should commence before the endorsement of RR(s) by EPD.

7.8.2 Chemicals of Concern and Possible Remediation Methods

7.8.2.1 With reference to the possible contaminants, remediation methods previously implemented in Hong Kong are presented in **Table 7.8**. If land contamination is identified, the actual remediation method shall be proposed in the Remediation

Action Plan for EPD endorsement. Remedial options are not limited to the remediation method listed in **Table 7.8**.

Table 7.8 Possible Remediation Methods for Potentially Contaminating Land Uses

Identified Potentially Contaminating Land Uses and Potentially Contaminating Activities	Chemicals of Concern (1)	Possible Remediation Methods (2)
Open area storage, container storage and warehouses where loading, unloading and storage of goods, fuel storage and transfer, maintenance of equipment and vehicles occurs.	Metals (full list), PCRs, VOCs and SVOCs.	Biopiling, Cement Solidification / Stabilisation, Permeable Reactive Barriers
Construction equipment storage / equipment depot and vehicle / repair maintenance workshops where release of oils, fuels and lubricants from vehicles, vehicle and equipment maintenance and refuelling may occur and where use of chemicals and solvents in maintenance activities and or motor vehicle painting and storage and disposal of wastes may have taken place.	Metals (e.g. chromium, copper, lead, manganese, nickel, zinc), PCRs, VOCs (e.g. acetone, BTEX, MTBE, and trichloroethene) and PAHs.	Biopiling, Cement Solidification / Stabilisation, Permeable Reactive Barriers
Factories/workshops where release of oils, fuels and lubricants from machinery, equipment maintenance and refuelling may occur and use of chemicals and solvents is possible.	Metals (full list), PCRs, VOCs and SVOCs.	Biopiling, Cement Solidification / Stabilisation, Permeable Reactive Barriers
Recycling facilities where storage and processing of waste materials, storage and transfer of chemicals and fuels. Storage and disposal of wastes.	Metals (full list), PCRs, VOCs, SVOC and PCBs (if applicable).	Biopiling, Cement Solidification / Stabilisation, Permeable Reactive Barriers, Thermal Desorption / Incineration

Note:

(1) Metals include antimony, arsenic, barium, cadmium, chromium III, chromium VI, cobalt, copper, lead, manganese, mercury, molybdenum, nickel, tin and zinc. Petroleum carbon ranges (PCRs) include C6 – C8, C9 – C16 and C17 – C35. Volatile organic compounds (VOCs) include BTEX (benzene, toluene, ethylbenzene, and total xylenes), MTBE (methyl tert-butyl ether), acetone, bromodichloromethane, 2-butanone, chloroform, methylene - chloride, styrene, tetrachloroethene, and trichloroethene. Semi-volatile organic compounds (SVOCs) include polycyclic aromatic hydrocarbons (PAHs) (acenaphthene, acenaphthylene, anthracene, benzo(a)anthracene, benzo(a)pyrene, benzo(b)fluoranthene, benzo(g,h,i)perylene, benzo(k)fluoranthene, chrysene, dibenzo(a,h)anthracene, fluoranthene, fluorene, indeno(1,2,3-cd)pyrene, naphthalene, phenanthrene and pyrene), bis-(2-ethylhexyl)phthalate, hexachlorobenzene, and phenol.PCBs - Polychlorinated biphenyls. BTEX – Benzene, Toluene, Ethylbenzene and total Xylenes

(2) Actual remediation method subject to site findings/condition and to be proposed in the RAP

7.8.2.2 According to Section **Table 7.6**, key chemicals of concern comprise:

- Volatile Organic Compounds (VOCs);
- Semi-volatile Organic Compounds (SVOCs);
- Metals;
- Petroleum Carbon Ranges (PCRs) and
- Polychlorinated biphenyls (PCBs).

7.8.2.3 According to the Practice Guide issued by EPD, the following factors should be considered when evaluating different remediation methods:

- Nature and level of contamination;
- Extent of contamination;
- Site characteristics (such as site hydrogeology, soil and groundwater chemical characteristics);
- Site constraints (such as available space, surrounding areas); and
- Time available for remediation.

7.8.2.4 The possibility of carrying out in situ remediation and recycling and reuse of remediated materials should be explored first. Ex situ remedial measures could then be considered if the in-situ remediation is not considered to be practical. Off-site disposal of contaminated materials to landfills should be adopted only as a last resort.

7.8.3 Remediation Methods for Soil and Groundwater Contamination

7.8.3.1 With reference to the possible contaminants, common remediation methods for contaminated soil are reviewed and provided in **Table 7.9**. These are just some remediation methods that have been practiced in Hong Kong. If land contamination is identified, the actual remediation method should be proposed in the RAP for EPD endorsement and is not limited to the remediation method listed in above tables.

Table 7.9 Possible Remediation Methods for Soil and Groundwater Contamination

Remediation Option	Target Contaminant	Description	Applicability / Advantages to Environment	Limitations / Disadvantages to Environment
Soil				
Biopiling	VOCs, SVOCs and PCRs	An Ex-situ bioremediation method that facilitates micro-organism growth in contaminated soil in order to degrade the waste into harmless products.	• Effective for treatment of soil contaminated with fuel hydrocarbons and non-halogenated VOCs. • Halogenated VOCs and SOVCs can also be treated. • Some successful local case studies, e.g. decontamination works at the Cheoy Lee Shipyard at Penny's Bay, reclamation works at North Tsing Yi Shipyard site, decommissioning of Kwai Chung Incinerator. • Cost-effective for treating large volume of contaminated soil • All materials and equipment are available in the market. • Can be designed to be a closed system and vapour emissions can be controlled	• Labour-intensive and considerable maintenance is required • Time-consuming (~12 months period) and not cost-effective for treating small volume of contaminated soil. • Biopile construction requires large area.
Bioventing	VOCs, SVOCs and PCRs	An In-situ bioremediation method which employs indigenous bacteria to degrade the contaminants. Bacterial activity is enhanced by air flow inducement (using extraction or injection wells) and by adding nutrients if necessary.	• Proven successful in remediating soil which is contaminated with PCRs and other organic compounds. • Suitable for remediation in built up areas (i.e. can be placed between and/or below buildings). • Applicable to extensive contamination in large sites. • All materials and equipment are available in the market and easy to install.	• Effectiveness may be affected by underground soil features e.g. soil moisture content, permeability, etc. • Potential air emission to the sensitive receivers System development requires large area.

Remediation Option	Target Contaminant	Description	Applicability / Advantages to Environment	Limitations / Disadvantages to Environment
			• Vapour emissions can be controlled but not to the extent of biopiling depending on underground soil in-situ properties.	
Soil Washing	Metals, SVOCs & PCRs	An Ex-situ soil separation method based on mineral processing techniques. A water-based process to remove contaminants by scrubbing soils ex-situ.	• Applicable to remove inorganic contaminants such as heavy metal from coarse-grained soils.	• Effectiveness dependent on soil coarseness. Addition of polymer may be required to remove contaminants. • The difficulty of formulating washing fluid increases with the complexity of waste mixtures. • Further treatment and disposal may be required for the residuals.
Solidification / Stabilization	Metals	An Ex-situ immobilisation technique which mixes contaminated soil with binding agents (e.g. cement) so as to physically bind contaminants into stable mass.	• Applicable to treat inorganic contaminants such as heavy metals. • Some successful local case studies, e.g. decontamination works at the Cheoy Lee Shipyard at Penny's Bay, reclamation works at North Tsing Yi Shipyard site, decommissioning of Kwai Chung Incinerator, and few isolated sites identified in the Deep Bay Link project.	• Not effective in reducing organic contaminants • Hindrance of large boulders in mixing process may be possible. Soil sorting is necessary before treatment.
Thermal Desorption	VOCs, SVOCs, PCRs and PCBs	A method to increase the volatility of contaminants by applying heat to the contaminated soil so that the contaminants can be removed / separated from the soil matrix.	• Low temperature thermal desorption systems are targeted for soils which are contaminated with VOCs, SVOCs and PCRs. • PAHs and pesticides can also be treated with high temperature thermal desorption. • Less polluting method relative to incineration.	• Effectiveness varies with different organic contaminants. • Processor unit can potentially be venerable to highly abrasive feed. • Reaction time varies with feed content as a result of binding of contaminants.

Remediation Option	Target Contaminant	Description	Applicability / Advantages to Environment	Limitations / Disadvantages to Environment
Incineration	VOCs, SVOCs, PCRs and PCBs	An Ex-situ or In-situ treatment of contaminated soils to break down contaminants by and hazardous materials by incinerating in order to break down contaminants.	• Applicable to soil contaminated with hazardous wastes such as chlorinated hydrocarbons. • Wide experience in Hong Kong. • Common practice for highly contaminated soils.	• Ash residues which contain volatile heavy metals are produced in the process. • Ash residues leaving the combustion unit require gas treatment before releasing to atmosphere. • Contaminants can react to each other to form more volatile and toxic compounds that further treatment may be required.
Groundwater				
Vapour extraction/ Groundwater extraction	PCRs	Vapour phase and residual hydrocarbons are recovered by applying vacuum to well(s). Free phase product and/or groundwater is recovered by pumping from the wells.	• Applicable to groundwater contaminated with VOCs, metals, PCBs and radioactive contaminants. • Often used to reduce chlorinated contaminants such as TCE and PCE. Effective method to immobilise metals such as uranium, chromium and arsenic.	• Water and vapour treatment and phase separation are typically required. • Startup time is longer than other conventional methods. • Moderate to large capital investment; medium to high operation and maintenance costs.
Permeable Reactive Barriers	Metals, VOCs, SVOCs, PCRs and PCBs	An In-situ treatment of contaminated water by means of engineering and use of reactive materials.	• Applicable to low permeability materials (clay, clayey silts, silts, silty / clayey sands). Surface seal (either naturally occurring clay or man-made) is required to prevent short-circuiting of vacuum. • Groundwater and product recovery increases with high vacuum. • Minimises drawdown and “smearing” of product. • Hastens site cleanup by recovering all hydrocarbon phases.	• Method only applies to relatively shallow aquifers. • Effectiveness of the treatment is dependent on groundwater chemistry / flow rate, ground geochemistry, site topography, seasonal changes in hydrogeological conditions.

Remediation Option	Target Contaminant	Description	Applicability / Advantages to Environment	Limitations / Disadvantages to Environment
Multiphase Extraction	VOCs, SVOCs and PCRs	Both liquid and vapour are recovered from same well. Production of groundwater is minimised, and water table is stabilised.	• Applicable to settings in which the amount of free product is small and exists in permeable conduits such as utility bedding or buried underground open structures. • Low cost and simple operation and maintenance. • Effective for shallow groundwater level and soil excavation works.	• Vapour and groundwater treatment and phase separation are usually required. • Longer startup time and higher capital cost
Skimming systems	PCRs	Free product is recovered from a pit or trench without recovering groundwater	• Applicable to settings in which the amount of free product is small and exists in permeable conduits such as utility bedding or buried underground open structures. • Low cost and simple operation and maintenance. • Effective for shallow groundwater level and soil excavation works.	• Recovery rates depend on size of pit/trench • Frequent replacement of media • Manual adjustment is required
Chemical oxidation	Metals, VOCs, SVOC, and PCRs	An In-situ treatment of water which converts hazardous contaminants to non-hazardous or less toxic compounds that are more stable, less mobile, and/or inert chemically.	• Rapid destruction / degradation of contaminants (measurable reductions in weeks or months. • No significant wastes are produced (VOC off-gas is minimal) • Contaminated soil can also be treated.	• Very specialised contractors are required. • Handling of large quality of hazardous oxidising chemicals is required. • Some contaminants are resistant to oxidation. • Pilot tests to determine the correct oxidant, dosage and time scale are required.
Free product recovery with water table depression	PCRs	Both free product and groundwater are recovered from a pit or trench. Groundwater is pumped to create cone of	• Capture zone is created which enables hydraulic control of groundwater and product	• Treatment is usually required for the recovered fluids treatment.

Remediation Option	Target Contaminant	Description	Applicability / Advantages to Environment	Limitations / Disadvantages to Environment
		depression in water table to expand area of influenced.	• Product recovery rates are enhanced by water table depression, especially in high permeability formations • Recovered groundwater can be oxygenated and re-injected for bioremediation	• Product can be “smeared” across area of depression resulting in greater formation storage. • Higher permeability formations may require high pumping rates. • Well network design requires capture zone analysis.

7.8.4 Surmountability of the Potentially Contaminated Sites

7.8.4.1 Contamination issues within those sites/areas identified as potentially contaminated are not insurmountable when the following factors are considered:

- The sizes of the potential contaminated sites are relatively small;
- The extent of potential contamination are likely to be relatively localized; and
- Remediation methods available are well established with local remediation experience sufficient to deal with the nature of the possible contaminants.

7.8.4.2 Land used for warehousing of non-contaminating goods is unlikely to present any land contamination potential. For open storage sites, a large portion of their areas are typically used for goods storage with possibly smaller portion for potentially contaminating activities such as vehicle / equipment maintenance area and the associated chemical handling/storage. If the main types of goods stored on-site are not potential sources of contamination, land contamination, if indeed present would likely be localised.

7.8.4.3 No large-scale polluting installations / facilities such as oil depots and power plants are identified.

Chemicals of Concern identified are treatable using established physical, chemical, and biological techniques.

7.8.4.4 Based on the site appraisal, the potential COCs identified with the potential to be present include metals, VOCs, SVOCs, PCRs and PCBs. Remediation methods that have been demonstrated to effectively treat the aforementioned soil and groundwater contaminants include biological treatment and physical / chemical treatment. Details of the treatment methods are provided in **Section 7.8.2 - 7.8.3.1**.

Local remediation experience to deal with the nature of the possible contaminants

7.8.4.5 Soil contaminated with the abovementioned COCs had successfully been remediated in Hong Kong using proven remediation techniques. Notable local remediation projects include the following:

- Decommissioning of Kai Tak Airport North Apron;
- Decontamination works at the Cheoy Lee Shipyard;
- Reclamation works at North Tsing Yi Shipyard site; and
- Decommissioning of Kwai Chung Incinerator

7.8.5 Tentative Schedule of Submission

7.8.5.1 The land resumption programme has not been confirmed, and would not commence until 2027. Once the access to potentially contaminated sites has been granted or a site vacated, those sites shall be reappraised. The tentative schedule of submission for each site is provided in **Table 7.10**.

7.8.5.2 It should be noted that the tentative schedule is for general reference only and the actual time span will be subject to the individual site conditions, phasing of development packages and land resumption program.

Table 7.10 Tentative Schedule of Submission

Submission	Time Required
Further site appraisal and submission of supplementary CAP	8 weeks
Endorsement of supplementary CAP by EPD	4 weeks
Contracting site investigation works and mobilization of contractor	4 weeks
Site investigation works and laboratory testing	2 - 10 weeks or more (Depends on actual proposed site investigation works)
If land contamination is not identified	
Preparation and submission of CAR	4 weeks
Endorsement of CAR by EPD	4 weeks
If land contamination is confirmed by CAR	
Preparation and submission of CAR/RAP	4 weeks
Endorsement of CAR/RAP by EPD	4 weeks
Remediation works and verification of result Preparation and submission of RR	12 weeks to 36 weeks (Length of remediation depends on the remediation method)
Endorsement of RR by EPD	4 weeks

7.9 Evaluation of Residual Impacts

7.9.1.1 The recommended further land contamination assessment and, if required, remediation works for the Project Site would follow the EPD's Guidance Manual, Guidance Note and Practice Guide and any soil / groundwater contamination would be identified and properly remediated prior to the commencement of construction works. Land contamination impacts are therefore considered acceptable if the follow up actions as outlined in **Section 7.8** are followed, and contaminated soil and groundwater (if any) are properly remediated using appropriate remediation methods and according to the EPD approved RAP(s). No unacceptable residual impacts are anticipated.

7.10 Environmental Monitoring and Audit

7.10.1.1 Remediation works, if necessary, would be carried out based on the recommended further works outlined in **Section 7.8**. Mitigation measures as recommended in the future EPD approved RAP(s) should be implemented during the remediation works. The Environmental Monitoring and Audit (EM&A) requirements should be carried out in the form of weekly site inspection to ensure the recommended mitigation measures are properly implemented.

7.11 Conclusion

7.11.1.1 Site appraisals, in the form of desktop review and site walkovers, had been carried out between September 2025 and January 2026 to identify the past and current potentially contaminating land uses within the Project Site. Based on the site appraisal, 30 potentially contaminated sites and 15 sites suspected to be used for industrial purposes were identified within the Project Site.

7.11.1.2 Further land contamination assessments, including further site appraisal and submission of CAP(s), should be conducted for the whole Project Site at a later stage of the Project when site access is available to confirm the existing land uses / activities, identify the presence of any potential contamination sources, and address any new contamination issues. The associated SI works and any necessary remediation action are recommended to be carried out after operation of concerned site(s) has ceased but prior to the commencement of construction works. The recommended further assessment and remediation works, including the submission of CAP(s), CAR(s)/RAP(s) and RR(s) would follow the relevant Guidance Manual, Guidance Note and Practice Guide.

7.11.1.3 With reference to previous studies, natural occurrences of elevated level of arsenic exist over broad areas of the northern New Territories. A separate arsenic assessment and remediation plan should be prepared and submitted to EPD for endorsement prior to the commencement of construction works.

7.11.1.4 With the implementation of the recommended follow up works for the Project, any soil/groundwater contamination would be identified and properly remediated prior to the construction works. No insurmountable land contamination impacts to the Project are therefore anticipated.